Artificial urinary sphincter surgery

Presenter script: the sixty minutes, slide by slide

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This is the spoken script for `presentation/AUS-teaching-deck.pptx` and the animated `AUS-teaching-deck-3d.pptx`. The slides carry no speaker notes (only picture and clip credits on the image slides), so this document is the script: print it, or keep it open on a second screen. The four clinical cases and every patient in the examples are fictional. Evidence statements follow the 5 October 2026 audit and the ST5 presenter guide; nothing new is claimed here.

How to read each slide

- **This slide wants:** the one thing the slide is asking the room to do.
- **On the slide:** what is actually shown.
- **Say:** the narration, in the "we" voice agreed on 10 October. Pauses are in square brackets.
- **Example:** one concrete, fictional example of the slide's ask, to make it land or to answer "what does that look like?"
- **Studies on this slide:** every citation code in the slide footer, what the source is, and what it does for this slide, with its limit. The full "what it says" for every code is in the glossary at the end.
- **Next:** the sentence that carries the room to the next slide.

The sixty minutes at a glance

Section	Teaching slides	Clock	Minutes
Decide before theatre	01–09	00:00 – 14:00	14
The bladder	10–14	14:00 – 22:00	8
The urethra	15–19	22:00 – 29:00	7
Preparation without ritual	20–24	29:00 – 36:00	7
The operation	25–32	36:00 – 50:00	14
After the implant	33–38	50:00 – 60:00	10

Before you start

- Add the meeting and date to the cover and your personal disclosures to teaching slide 02.
- Open https://uroops3d.com/lab/aus in a second window before the session for teaching slide 32. If it fails, the slide's notes carry the four-minute spoken fallback, and the animated deck has a recorded run-through as a backup slide after the discussion.
- The slides carry no timings. The clock beside each slide in this script is the running time at which that slide should start; the six section slides take no time.
- Say aloud, when the first UroOps picture appears, that the model is a teaching schematic with clinical review pending.
- Examples are there to be used if the room needs one. Skipping an example saves about twenty seconds; the clock already allows for the Say text alone.

Decide before theatre

00:00 – 14:00 · 14 minutes · teaching slides 01 to 09

Section slide, no clock time. Read the title, then say: "Can this patient obtain useful continence while storing and emptying safely and operating the device?"

01 The decision before theatre

00:00 – 01:30 · 1:30 · Decide before theatre

This slide wants: Sort every AUS candidate into one of four answers: implant, optimise, investigate or reconsider.

On the slide: The decision lens for the whole hour: implant, optimise, investigate or reconsider.

SAY

An AUS can be a technically satisfactory implant and a disappointing treatment. The cuff may coapt, the pump may cycle, and the patient may still avoid leaving the house because urgency, difficult emptying or fear of another procedure dominates his life.

Today we want to follow the decisions that separate those outcomes. We will start with what the patient actually wants, then examine the bladder and urethra that are to be operated on. We will spend time on evidence-based preparation and the small operative decisions that are difficult to rescue later.

One question will keep coming back, and it is on the slide in four words. Would we implant now, optimise first, investigate further, or reconsider whether an AUS is the right treatment? Each of those choices needs a reason, and we will keep returning to them with four fictional patients.

EXAMPLE

A 68-year-old uses five pads a day after prostatectomy. Stress leakage is obvious on coughing, but he also voids every hour with urgency. The four-word question sorts him: not "implant" yet, because a treatable bladder problem may dominate; "optimise" first, then reassess. The same four words will be asked of every patient today.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **BAUS25** — BAUS consensus 2025 (Bhatt et al., BJU Int) (UK expert consensus after literature review and modified Delphi). Its pathway is the four-word question in guideline form: assess storage and emptying, treat narrowing first, confirm device suitability. It supports a consistent pathway, not a validated score.
- **AUA24** — AUA/GURS/SUFU incontinence after prostate treatment, amended 2024 (US guideline). Supports implanting only for bothersome, demonstrated SUI with a patient able to manage the device; its counselling and pump-suitability statements are the "reconsider" column.
- **ICS16** — ICS AUS consensus 2015, published 2016 (Biardeau et al.) (International consensus, centred on AMS 800). Lists adequate dexterity and cognition plus bladder and outlet evaluation as prerequisites: the "implant" column's conditions. AMS 800-era consensus, not a ContiClassic rule.

Next → Before we use any of that evidence, we should be honest about what kind of evidence it is.

02 Evidence and device boundaries

01:30 – 03:00 · 1:30 · Decide before theatre

This slide wants: Name what kind of evidence sits behind each claim before leaning on it.

On the slide: The four kinds of authority we will cite, and the boundary between devices.

SAY

Thank you to Rigicon for organising the meeting. The clinical discussion needs to stay independent of the sponsor, so device choice should follow the evidence, patient suitability and the instructions for that device.

We have useful guidance, but agreement among experts is not the same as a comparative trial. A manufacturer's instruction tells us how its product is intended to be used. It tells us nothing about whether that product outperforms another. A retrospective series can identify a concern without proving that the proposed solution prevents it.

The BAUS consensus is particularly useful because it sets out the clinical pathway. We will compare it with EAU and AUA guidance where they differ. The ICS and Asia-Pacific consensus documents are also useful, but both are centred on the AMS 800.

Our aims are to choose investigations sensibly, separate three different bladder problems, preserve a usable urethra, question preparation rituals and diagnose failure before choosing revision. Where the evidence stops, we will say so.

EXAMPLE

Someone says "the guideline requires fifteen minutes of skin preparation". Ask which source that is. It turns out to be local habit, not BAUS, not NICE and not the IFU. That is an "expert or local practice" row: a practical choice with stated uncertainty, not a rule.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **BAUS25** — BAUS consensus 2025 (Bhatt et al., BJU Int) (UK expert consensus after literature review and modified Delphi). The example of "guidelines and consensus": a UK pathway built by literature review and Delphi, which supports a recommended pathway but proves no single step optimal.
- **EAU26** — EAU non-neurogenic male LUTS guideline 2026 (European guideline). A second guideline whose positions differ from BAUS (weak recommendation for urodynamics), which is why the slide says the sources must be compared, not merged.
- **AUA24** — AUA/GURS/SUFU incontinence after prostate treatment, amended 2024 (US guideline). The third guideline; its expert-opinion cystoscopy statement and conditional urodynamics show that guideline strength varies row by row.
- **ICS16** — ICS AUS consensus 2015, published 2016 (Biardeau et al.) (International consensus, centred on AMS 800). "Clinical consensus" centred on the AMS 800: useful principles, but its technical advice belongs to that device and era.
- **AP23** — Asia-Pacific AMS 800 consensus 2023 (Chung et al.) (Regional expert consensus). A regional AMS 800 consensus whose pressures and preparation sequence illustrate the boundary between devices: AMS 800 numbers do not transfer to ContiClassic.
- **CCIFU** — Rigicon ContiClassic instructions for use, REV.03, 2023 (Manufacturer labelling). The "manufacturer IFU" row: permitted use and precautions for ContiClassic, which is product instruction, not proof of efficacy or superiority.

- **AUASTATUS26** — AUA current guideline register, July 2026 (Official register page). Confirms the AUA guideline is still the 2024 amendment, so 2026 meeting abstracts are separate evidence, as the footer says.
- **DEVICE26** — Emerging AUS devices review 2026 (Schnell et al., Neurourol Urodyn) (Narrative review). A narrative review of new devices that stresses the lack of direct comparisons: the reason device results are "not interchangeable".

Next → So let us begin where the patient begins: with the leak itself.

03 Leakage mechanism and patient goals

03:00 – 04:45 · 1:45 · Decide before theatre

This slide wants: Describe the leak precisely and agree the improvement the patient actually wants.

On the slide: Characterising the leak and agreeing what improvement would be worth an implant.

SAY

Start with a description of a bad day. Does he leak when standing, walking or coughing? Does an urge come before the leak? Is he dry overnight, or does night-time leakage suggest another cause? Mixed symptoms are common, but the word mixed must not replace an explanation.

Demonstrate stress leakage with a suitably filled bladder and a practical provocative test. If the history and examination disagree, investigate that disagreement. A dry examination just after he has emptied his bladder proves very little.

Pad number is easy to record, but it depends on pad capacity and the patient's changing behaviour. The 2026 pad study tells us that count tracks leakage well overall, yet two to four pads a day hides a wide range of weights. A pad-weight assessment quantifies leakage when severity or progress is uncertain, provided the collection conditions and activity are recorded. A diary helps separate urinary frequency from precautionary voiding.

Then ask what improvement would justify an implant. Walking the golf course, returning to work or managing with one precautionary pad may matter more than a label of continence. Record a validated symptom and quality-of-life measure, and keep the patient's own goal beside it.

EXAMPLE

A man reports three pads a day. A 24-hour pad weight shows 450 g, far more than his count suggested. His goal is to walk the dog without changing a pad halfway. That goal, with the weight beside it, is what the implant has to deliver.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **BAUS25** — BAUS consensus 2025 (Bhatt et al., BJU Int) (UK expert consensus after literature review and modified Delphi). Asks for a structured history, demonstrated stress leakage, pad assessment and a validated questionnaire before surgery: the left column in guideline form.
- **EAU26** — EAU non-neurogenic male LUTS guideline 2026 (European guideline). Recommends diary, pad-weight and residual measurement as structured assessment: the measures listed under "characterise the leak".

- **AUA24** — AUA/GURS/SUFU incontinence after prostate treatment, amended 2024 (US guideline). Supports defining bother and improvement with the patient and documenting a validated measure before offering surgery: the right column.
- **PAD26** — Pad count versus 24-hour pad weight 2026 (Domínguez Argomedo et al., BJU Int) (Retrospective analysis of prospective follow-up, 2,040 men, 8,787 paired measurements). Shows pad count tracks leakage well overall (rho 0.94) but hides wide weight variability at two to four pads a day, which is exactly the footer line: measure weight when severity is uncertain.

Next → Once we know what he wants back, the next question is when, and whether an AUS is the only route there.

04 Timing after prostate treatment

04:45 – 06:15 · 1:30 · Decide before theatre

This slide wants: Pick the moment to operate from the recovery curve, not the calendar, and say where a sling fits.

On the slide: When to operate after prostate treatment, and where the sling trial fits.

SAY

Do not let conservative management drift on indefinitely. But do not interrupt a convincing recovery without a reason either. Establish what has been tried, whether pelvic-floor training was properly taught, and whether leakage is still improving.

BAUS allows a recovery period of at least six months for severe leakage and twelve for mild-to-moderate leakage. Severity here is the 24-hour pad weight on the Nitti scale, which BAUS adopts: mild under 100 grams, moderate 100 to 400, severe over 400. The AUA permits surgery from six months when bothersome stress leakage is not improving despite conservative treatment, and recommends offering it at one year when it persists. Anticipated salvage treatment, recurrence and the patient's recovery so far still belong in the decision.

A sling remains an option for selected men with less severe leakage and favourable tissues. Severe leakage and previous radiation generally move the discussion towards AUS. The AUA specifically favours AUS over slings or adjustable balloons in irradiated men seeking surgery. Containment, collecting devices and continued conservative care remain legitimate choices.

The MASTER trial supports an informed comparison. At 24 months the sling remained non-inferior on its primary continence endpoint, but further surgery was needed in eleven per cent after sling against two per cent after AUS. Equal on one measure does not mean equal for every patient. The trial excluded men with a previous implant or an untreated narrowing, so its result does not apply to complex or revision cases.

EXAMPLE

Eight months after prostatectomy a man still uses four pads a day, has done supervised pelvic-floor training and has not improved for three months. Under BAUS he is past the six-month recovery period for severe leakage; under AUA, surgery may be offered. He has no radiotherapy history, so a sling is a legitimate alternative to discuss. If he had been irradiated, the AUA would favour AUS.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **AUA24** — AUA/GURS/SUFU incontinence after prostate treatment, amended 2024 (US guideline). The AUA row: surgery may be offered from six months for bothersome, non-improving SUI and should be offered at one year; it also favours AUS over sling or balloons in irradiated men.
- **BAUS25** — BAUS consensus 2025 (Bhatt et al., BJU Int) (UK expert consensus after literature review and modified Delphi). The BAUS row and the severity strip: at least six months' recovery for severe SUI and twelve for mild-to-moderate, as consensus, not as a trial-proven interval. Severity is the 24-hour pad weight on the Nitti scale (statement 9): mild under 100 g, moderate 100 to 400 g, severe over 400 g.
- **EAU26** — EAU non-neurogenic male LUTS guideline 2026 (European guideline). Supports conservative management first and a structured reassessment of the recovery trajectory before invasive treatment.
- **MASTER22** — MASTER trial 2022 (Constable et al., NIHR HTA) (UK multicentre randomised non-inferiority trial, 380 men, sling versus AUS). The randomised comparison behind "where a sling fits": at twelve months the sling met non-inferiority on patient-reported continence in 380 selected men with no prior implant or unresolved narrowing. The further surgery at twelve months was mostly an AUS after a failed sling (12 of the 13 sling-arm reoperations; one had a second sling), while the three AUS-arm reoperations were revisions for mechanical failure.
- **MASTER26** — MASTER 24-month results 2026 (Constable et al., Eur Urol Focus) (Extended follow-up of the same 380 randomised men). The footer: at 24 months the sling remained non-inferior, but further surgery was 11% after sling versus 2% after AUS, and secondary outcomes favoured AUS. Non-inferiority on one endpoint is not interchangeability.

Next → Timing assumes the patient can use the device. That is the next thing to test, and we test it rather than ask about it.

05 Pump use over time

06:15 – 07:30 · 1:15 · Decide before theatre

This slide wants: Watch the patient operate a pump and judge whether he can still do it in five years.

On the slide: A demonstrated task: can this patient operate this pump now and in five years?

SAY

Ask the patient to operate a demonstration pump, and watch what happens. Arthritis, neuropathy, tremor and weak pinch may only show when he attempts the task.

Access matters as much as strength. Can he reach the proposed side of the scrotum while seated? Obesity, a high scrotum, limited hip movement or another scrotal device can make a well placed pump unusable. Agree the side with him, because his dominant hand alone does not settle it.

Check that he understands cycling, deactivation and what to tell another clinician before catheterisation. Cognitive decline and progressive disease change the long-term plan. Family support helps, but it does not automatically help a man who cannot operate the device when he is alone.

Frailty is more than an age cutoff. The 2026 Medicare cohort links frailty to early complications and to later revision, independently of age. Frailty there was scored from billing codes, the Claims-Based Frailty Index, not measured at the bedside, so it tells us frailty matters without telling us how to test it in clinic. Balance anaesthetic burden, recovery,

expected benefit and the likelihood of future revision. If safe independent use is unlikely, reconsider the treatment.

EXAMPLE

Hand the patient a demonstration pump in clinic. A man with rheumatoid hands can describe the device perfectly but cannot squeeze it three times in a row. That observation, not his age or his answer, is what moves him from "implant" towards "reconsider" or towards a carer-assisted plan.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **AUA24** — AUA/GURS/SUFU incontinence after prostate treatment, amended 2024 (US guideline). States that candidates must be able to operate the device and that this is part of counselling: the reason the pump is tested, not asked about.
- **ICS16** — ICS AUS consensus 2015, published 2016 (Biardeau et al.) (International consensus, centred on AMS 800). Names adequate dexterity and cognition as prerequisites for an AUS, which is what the demonstration pump checks.
- **CCIFU** — Rigicon ContiClassic instructions for use, REV.03, 2023 (Manufacturer labelling). Labels the device as unsuitable for a patient who cannot operate the pump, and specifies deactivation, which the patient must understand.
- **FRAIL26** — Frailty and SUI surgery 2026 (Nik-Ahd et al., J Urol) (Medicare claims cohort, 7,252 men, AUS and sling). The "frailty as well as pump use" bullet: in 7,252 men, claims-defined frailty raised 30-day complications (adjusted RR 1.5 prefrail, 2.5 frail) and revision, independently of age. A claims index, so it validates no bedside score.

Next → With the patient assessed, we turn to the two investigations that most often change the plan. First the cystoscope.

06 Flexible cystoscopy before implantation

07:30 – 09:15 · 1:45 · Decide before theatre

This slide wants: Decide what a cystoscopy is for, and which findings would change the plan.

On the slide: What a flexible cystoscopy is for, and which findings change the plan.

SAY

The AUA recommends cystourethroscopy before stress-incontinence surgery. BAUS has two routes. Where video-urodynamics is available, every surgical candidate has VUDS, and cystoscopy is added only when something points to a narrowing: voiding symptoms, a poor flow or residual, obstruction or a suspicious outline on the screening film, or blood on the dipstick. Where video is not available, every candidate has standard urodynamics plus cystoscopy, because the scope replaces the X-ray view of the bladder neck, sphincter and urethra. BAUS also allows an on-table cystoscopy before the incision if patency is in doubt. EAU describes selective cystoscopy for narrowing or bladder pathology, with a weak general recommendation when the findings would change diagnosis or treatment.

Our practical preference before an AUS is to inspect the urethra and bladder in advance, so that an unexpected finding can be discussed before the day of implantation. That is our preference, because the guidelines do not agree. Repeat the assessment after any urethral treatment or new obstructive symptoms.

Look for a stricture, vesicourethral anastomotic stenosis, bladder-neck contracture, foreign material or previous cuff erosion. Bladder stones, tumour and significant radiation changes may alter treatment or predict repeated instrumentation.

With suspected narrowing, add flow and residual assessment and define the site and length. Retrograde urethrography, sometimes with a voiding study, is useful when reconstruction is being considered or the segment cannot be assessed endoscopically. And do not turn a diagnostic cystoscopy into an unplanned dilation simply to complete the examination. A dilation is a treatment: it resets BAUS's six-month stability clock, it tears the scar before we have recorded its site, length and calibre, so the imaging and the reconstructive plan that follow are less reliable, and each tear adds spongiofibrosis to the very tissue a cuff may need. If the narrowing needs treating, that is a planned, consented decision about which treatment.

EXAMPLE

Routine flexible cystoscopy before an AUS shows a 1 cm bladder-neck contracture that a 16 Fr scope barely passes. The implant is postponed, the contracture is treated, and stability is reassessed. The cystoscopy did its job: it changed the plan before the day of implantation.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **BAUS25** — BAUS consensus 2025 (Bhatt et al., BJU Int) (UK expert consensus after literature review and modified Delphi). The BAUS row: VUDS for every surgical candidate; cystourethroscopy only in selected cases (suspected bladder-neck stenosis or stricture, or dipstick haematuria); where VUDS is unavailable, standard urodynamics plus cystourethroscopy for all; and an on-table cystoscopy before the incision if patency is in doubt.
- **AUA24** — AUA/GURS/SUFU incontinence after prostate treatment, amended 2024 (US guideline). The AUA row: cystourethroscopy before SUI surgery is recommended, as expert opinion; it also supports interval cystoscopy after treating BNC or VUAS.
- **EAU26** — EAU non-neurogenic male LUTS guideline 2026 (European guideline). Describes selective cystoscopy for suspected narrowing or bladder pathology with a weak general recommendation: the "practical preference" framing.
- **EAUSTR26** — EAU urethral strictures guideline 2026 (European guideline). Gives the anatomical definitions of stricture, bladder-neck contracture and VUAS that the "findings that change the plan" column uses, and the imaging to define site and length.
- **AUAUDS26** — AUA 2026 abstract IP09-22: pre-AUS urodynamics and cystoscopy (Conference abstract, title and programme verified only). The footer: a 2026 abstract revisiting preoperative testing whose title and programme were verified but whose results were unavailable, so nothing changes yet.

Next → The second investigation is where the guidelines genuinely disagree, so we will name the positions rather than blur them.

07 Urodynamics: BAUS, EAU and AUA

09:15 – 11:00 · 1:45 · Decide before theatre

This slide wants: State the urodynamic question before ordering the test, knowing the three guidelines differ.

On the slide: Three guidelines, three positions on urodynamics. We say which one we follow.

SAY

BAUS favours video-urodynamics in the surgical pathway. EAU recommends urodynamics before invasive treatment, with weak strength, and its narrative still discusses selective testing and limited outcome evidence. The AUA permits testing when it helps diagnosis or counselling. The older ICS consensus also takes a selective approach.

The yield is greatest when stress leakage is uncertain, urgency is dominant, emptying is poor, compliance is a concern, or previous treatment has complicated the mechanism. In an uncomplicated man with demonstrable severe stress leakage, reassuring emptying and no worrying storage symptoms, urodynamics may add little under the AUA approach. BAUS and EAU still take a different position.

Before requesting the test, state the question. Is this unsafe storage, obstruction, weak contraction, or a bladder that never gets a chance to fill? Decide in advance what would change surgery or counselling.

We found no comparative trial showing that routine testing itself improves AUS outcomes for every candidate. A consensus can recommend a sensible investigation without proof that its universal use is superior. The 2026 AUA abstract revisits this question, but its full results were not available, so we quote the topic and nothing more.

EXAMPLE

Two colleagues disagree about ordering urodynamics for an uncomplicated man with demonstrable severe stress leakage, good emptying and no urgency. One follows BAUS (video-urodynamics for surgical candidates); the other follows AUA (test only when it changes diagnosis or counselling). Both are following a guideline. The honest answer is to name the guideline you follow and the uncertainty the test would resolve.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **BAUS25** — BAUS consensus 2025 (Bhatt et al., BJU Int) (UK expert consensus after literature review and modified Delphi). The BAUS row: video-urodynamics for surgical candidates as the investigation pathway.
- **EAU26** — EAU non-neurogenic male LUTS guideline 2026 (European guideline). The EAU row: urodynamics before invasive treatment as a weak recommendation, with a narrative that still discusses selective testing.
- **AUA24** — AUA/GURS/SUFU incontinence after prostate treatment, amended 2024 (US guideline). The AUA row: urodynamics conditional (Grade C) when it helps diagnosis or counselling, so the clinician must state what the test should resolve.
- **ICS16** — ICS AUS consensus 2015, published 2016 (Biardeau et al.) (International consensus, centred on AMS 800). Took a selective approach to urodynamics before an AUS, showing the disagreement is not new.
- **AUASTATUS26** — AUA current guideline register, July 2026 (Official register page). Confirms that the AUA position quoted is still current (amended 2024, no 2026 amendment).
- **AUAUDS26** — AUA 2026 abstract IP09-22: pre-AUS urodynamics and cystoscopy (Conference abstract, title and programme verified only). The footer: conference material on this exact question, results unverified, quoted as a topic and nothing more.

Next → Whichever pathway you follow, the study you get back will be imperfect. Here is how we read it.

08 Interpreting an imperfect study

11:00 – 12:45 · 1:45 · Decide before theatre

This slide wants: Read an imperfect urodynamic trace beside the real patient, in five steps.

On the slide: A five-step way to read an imperfect urodynamic study beside the real patient.

SAY

A man who leaks throughout filling may never reach a meaningful storage challenge. A small recorded capacity can reflect leakage, urgency, discomfort, filling conditions or genuinely poor storage, and each of those leads to a different decision.

Review the actual traces, pressure subtraction, filling rate, leakage and the reason filling stopped. A compliance value calculated during artefact or substantial unrecorded leakage carries no authority just because the machine printed it. Compare it with his diary and usual voided volumes.

A catheter in place during a pressure-flow study may inhibit voiding or alter outlet behaviour. Weissbart's series is the reminder: thirty-two of 169 men did not leak with the catheter in and every one of them leaked once it came out. And a man who empties by abdominal straining through a very incompetent outlet may not behave the same after continence surgery. Repeat free flow and residual measurements under representative conditions, and explain the remaining uncertainty.

The pathway runs like this. Establish the dominant symptom and demonstrable stress leakage. Assess urine, diary, flow and residual, and evaluate the urethral anatomy. Use the chosen guideline-based urodynamic strategy, then classify storage safety, emptying and outlet stability. If a study is technically inadequate and its result would decide whether implantation is safe, investigate further. An uninterpretable trace is neither a contraindication nor reassurance.

EXAMPLE

A trace reports capacity 180 mL and compliance 12 mL/cmH2O. Looking at it, the patient leaked continuously from 100 mL, so the bladder never filled. His diary shows voided volumes of 350 mL. The printed numbers are artefacts of leakage, not a small, stiff bladder. The decision is "investigate", not "reconsider".

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **BAUS25** — BAUS consensus 2025 (Bhatt et al., BJU Int) (UK expert consensus after literature review and modified Delphi). Its pathway supplies steps one to five: symptoms and demonstrated SUI, urine and flow, urethral anatomy, the urodynamic strategy, then classification of storage and emptying.
- **AUA24** — AUA/GURS/SUFU incontinence after prostate treatment, amended 2024 (US guideline). Supports repeating free flow and residual under representative conditions and interpreting urodynamics against counselling needs, not in isolation.
- **ICS16** — ICS AUS consensus 2015, published 2016 (Biardeau et al.) (International consensus, centred on AMS 800). Advises that a technically poor or uninterpretable study should prompt further investigation rather than a decision either way.

- **UDS14** — Weissbart et al. 2014: absent leak during intubated urodynamics (Cohort, 169 men). The reason for "a weak laboratory contraction needs the real voiding history": 32 of 169 men did not leak with the test catheter in and every one leaked when it came out. A catheter can mask the real picture.

Next → When the assessment is complete, we consent. And consent is where the numbers need their denominators.

09 Consent: improvement and lifetime burden

12:45 – 14:00 · 1:15 · Decide before theatre

This slide wants: Quote outcome numbers with their population, endpoint and time horizon.

On the slide: Two honest numbers, each with its population, endpoint and time horizon.

SAY

Consent includes residual leakage, urgency, erosion, infection, mechanical failure and revision. Radiation and previous surgery change those risks, and future instrumentation needs precautions.

AUSCO gives contemporary AMS 800 context: 115 primary implants, including 25 previously irradiated men. At twelve months after activation, 61 of 101 reporting participants used zero pads. It was a manufacturer-funded, single-arm study in expert centres, and missing outcomes and one-year follow-up limit what we can infer.

The first 116-recipient ContiClassic series reported 93.2 per cent Kaplan–Meier device survival at twelve months. A standardised continence result and primary, revision or radiation subgroup outcomes could not be verified in the accessible abstract. That tells us about early device survival. It does not tell us about pad freedom, and it makes no comparison between devices.

Quote each endpoint with its population and time horizon. The patient should know what the device may improve and what happens if it has to be removed. The 2026 regret study makes that concrete: reoperation was the only independent predictor of regret, and benefit faded with time, so future surgery belongs in the conversation from the start.

EXAMPLE

The patient asks "what are my chances of being dry?" The answer has three parts: in AUSCO, 61 of 101 men used zero pads at twelve months after activation with an AMS 800; the ContiClassic series reports 93.2% device survival at twelve months, which is not continence; and the regret data say reoperation, not residual leakage, is what men later regret. Three numbers, three populations, three endpoints.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **AUSCO26** — AUSCO study 2026 (Kaufman et al.): prospective AMS 800 outcomes (Prospective single-arm, manufacturer-funded, 17 sites, 115 primary implants). The left number: 61 of 101 reporting men used zero pads twelve months after activation; 115 primary AMS 800 implants, single-arm, manufacturer-funded, with missing denominators.
- **CCSAFETY** — Wilson et al.: first ContiClassic safety outcomes, IJIR 2024 (Uncontrolled early device series, 116 implants, mixed indications). The right number: 93.2% twelve-month Kaplan–Meier device survival in the first 116 ContiClassic recipients; device survival, not continence, and no comparator.

- **AUA24** — AUA/GURS/SUFU incontinence after prostate treatment, amended 2024 (US guideline). Requires lifetime counselling about erosion, infection, mechanical failure and revision: the "lifetime burden" in the title.
- **RT22** — Zhang and Xu 2022: radiation meta-analysis (Meta-analysis of 18 observational studies). Supplies the radiation caveat for consent: pooled odds of erosion or explantation roughly doubled (OR about 2.0) in irradiated men.
- **FRAIL26** — Frailty and SUI surgery 2026 (Nik-Ahd et al., J Urol) (Medicare claims cohort, 7,252 men, AUS and sling). Adds frailty-related early complications and later revision to the consent conversation.
- **REGRET26** — Decisional regret after AUS 2026 (Roth et al., Eur Urol Open Sci) (Retrospective single-centre survivor cohort, 91 respondents, median follow-up 82 months). The footer's "long-term regret data": reoperation was the only independent predictor of regret (+18 points), and benefit faded over 82 months. Future surgery belongs in consent from the start.
- **DEVICE26** — Emerging AUS devices review 2026 (Schnell et al., Neurourol Urodyn) (Narrative review). Reminds us that the two numbers come from different devices with no direct comparison, which is why "these endpoints do not compare devices".
- **REPORT26** — Standardised complication reporting 2026 (Roessler et al., World J Urol) (Retrospective series, 227 implants, systematic complication capture). Shows what systematic reporting looks like: 47% early complications, mostly minor, 7.9% major, 73% five-year explantation-free survival in a mixed-technique series. It explains why similar-looking percentages differ.

Next → That is the decision before theatre. Now let us put it to work with our first fictional patient, and hand the decision to you.

The bladder

14:00 – 22:00 · 8 minutes · teaching slides 10 to 14

Section slide, no clock time. Read the title, then say: "Separate overactivity, weak contraction and poor compliance, because each leads to a different decision."

10 Case 1: six pads and urgency

14:00 – 15:45 · 1:45 · The bladder

This slide wants: Decide, for a mixed-leak patient, whether to implant now or treat the urgency first.

On the slide: Case 1 tests the lens on a bladder that leaks two ways: stress and urgency.

SAY

This fictional patient uses six pads on an active day. Stress leakage is clearly demonstrated. He also has urgency leakage, and urodynamics shows detrusor overactivity, with reassuring storage pressures between contractions and adequate emptying. His urethra is patent.

Would you implant now or treat the bladder first? [Pause for eight seconds.] Both may be defensible in general AUS practice, depending on which symptom dominates, how severe the urgency is and what the device's instructions say.

Overactive bladder is a symptom syndrome, whereas detrusor overactivity is an observation during filling. One does not perfectly predict the other. A systematic review found insufficient grounds to deny stress-incontinence surgery solely because DO was present, but the studies and definitions were heterogeneous. It does not guarantee that urgency or device durability is unaffected; in one reintervention cohort, DO went with earlier reintervention.

Our approach would be to offer treatment for bothersome storage symptoms, agree a separate goal for urgency, and reassess. If stress leakage remains the dominant burden and storage is safe, AUS can remain appropriate. For ContiClassic specifically, the retrieved IFU requires these bladder conditions to be treated and controlled or resolved, and irresolvable instability is contraindicated. General reassurance about DO must not override that device boundary.

EXAMPLE

Case 1 is on the slide. A good answer sounds like this: "Optimise. Treat the urgency, agree that urgency is a separate outcome, and reassess in three months. If stress leakage still dominates and storage is safe, implant. For a ContiClassic I would also need the bladder condition controlled or resolved, because the IFU says so."

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **DO24** — Bhatt et al. 2024: preoperative DO systematic review (Systematic review, 11 studies, 792 patients, AUS and sling). The footer's first sentence: a systematic review of 792 patients found no significant excess of failure sufficient to deny surgery solely for DO. Heterogeneous definitions, so it supports counselling, not a promise.

- **OAB11** — Lai and Boone 2011: OAB and post-prostatectomy AUS (Selected cohort, 129 first AUS after prostatectomy). Warns that stress leakage can improve while urgency persists: 71% of mixed-symptom men still had OAB after AUS and 23% with pure SUI developed it.
- **DOFAIL23** — Krughoff and Peterson 2023: earlier AUS reintervention (Single-centre analysis of 92 men already undergoing reintervention). Tempers the reassurance: in 92 men already having reintervention, DO went with earlier reintervention (adjusted HR 1.95). A selected group, so no failure probability for primary recipients.
- **CCIFU** — Rigicon ContiClassic instructions for use, REV.03, 2023 (Manufacturer labelling). The footer's second sentence: ContiClassic labelling requires urge or instability to be treated and controlled or resolved, and contraindicates irresolvable instability. A device boundary that general reassurance cannot override.

Next → So how do we treat that urgency without creating the next problem, an emptying problem?

11 Urgency treatment and emptying

15:45 – 17:15 · 1:30 · The bladder

This slide wants: Treat urgency in an order that never creates an emptying problem behind a cuff.

On the slide: Treating urgency in the right order, with emptying and future access in view.

SAY

Begin with the reversible contributors: fluid pattern, caffeine, constipation, infection when clinically suspected, and medications. Bladder training and an appropriate antimuscarinic or beta-3 agonist may be useful, chosen with the patient's comorbidity, interactions and residual urine in mind. We are not prescribing a universal drug sequence.

Botulinum toxin can be effective for refractory urgency, but think through its consequences before an AUS. Could this patient catheterise if needed? How would repeated cystoscopic injections be delivered safely through the future cuff? If retention develops, we have exchanged one functional problem for another. Neuromodulation can also be considered through an appropriate overactive-bladder pathway. We still need to establish that the outlet is stable and emptying is adequate.

Return to the patient's two outcomes: stress leakage and urgency leakage. Lai's cohort tells us that most men with mixed symptoms still had urgency after a successful AUS. Improvement in one is not failure because the other persists, provided that possibility was properly discussed.

But do not dismiss severe postoperative urgency as an inevitable inconvenience. Reassess it, including infection, residual urine and device function, before escalating medication.

EXAMPLE

A patient with mixed leakage is started on a beta-3 agonist rather than an antimuscarinic because of early cognitive impairment. Botulinum toxin is not offered yet, because he could not self-catheterise if he went into retention and every repeat injection would have to pass a deactivated cuff. Urgency improves; stress leakage remains the dominant burden; the AUS discussion resumes.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **BAUS25** — BAUS consensus 2025 (Bhatt et al., BJU Int) (UK expert consensus after literature review and modified Delphi). Places treatment of bothersome storage symptoms and reassessment before continence surgery: steps one and four.
- **AUAOAB24** — AUA/SUFU idiopathic OAB guideline 2024 (Cameron et al.) (US guideline). The evidence-based menu for steps one to three: behavioural measures, antimuscarinic or beta-3 choice with cognition in mind, and the retention implications of botulinum toxin. It sets no sequence around an AUS.
- **OAB11** — Lai and Boone 2011: OAB and post-prostatectomy AUS (Selected cohort, 129 first AUS after prostatectomy). Step four's justification: after a successful AUS most men with mixed symptoms still had urgency, so urgency must be agreed as a separate outcome beforehand.
- **CCIFU** — Rigicon ContiClassic instructions for use, REV.03, 2023 (Manufacturer labelling). Requires bladder instability to be controlled or resolved before a ContiClassic, which is why the urgency plan comes before the implant date.

Next → Our second patient has the opposite bladder problem. He cannot empty well even before we add a cuff.

12 Case 2: weak contraction and residual urine

17:15 – 19:00 · 1:45 · The bladder

This slide wants: Decide whether a weak bladder with residual urine can safely take a cuff.

On the slide: Case 2 asks whether a weak bladder with residual urine can safely take a cuff.

SAY

Our second fictional patient has demonstrable stress leakage but voids slowly, strains and repeatedly leaves urine behind. The pressure-flow study shows a weak contraction. Is that enough to refuse him an AUS? [Pause for eight seconds.]

First exclude a narrowing, review medications and neurological history, and compare the laboratory void with his usual void. A large residual with a weak contraction still needs an explanation. A technically poor study should not define his future treatment.

Selected retrospective data suggest that men labelled as having detrusor underactivity can do reasonably well after AUS. But that study excluded early complications requiring revision, and the accessible report does not allow confident extrapolation to every patient with severe retention or an acontractile bladder.

An AUS opens during cycling, so it differs from a permanently obstructive outlet procedure. Even so, we cannot promise that a man who empties mainly by straining will empty equally well after implantation. Our initial decision is to investigate further and establish a practical emptying plan. If that becomes reliable, implantation may be reasonable. If he cannot empty or safely manage the likely rescue strategy, reconsider AUS. No universal residual or contractility number makes this decision for us.

EXAMPLE

Case 2 is on the slide. A good answer sounds like this: "Investigate. Exclude a stricture, repeat free flow and residual at home volumes, and agree an emptying plan. If he can reliably empty, or can safely self-catheterise through a deactivated cuff, implantation becomes reasonable. If neither is true, reconsider." Not a number, a plan.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **DU23** — Han et al. 2023: AUS with detrusor underactivity (Retrospective cohort, 78 patients, 55 with BCI below 100). The footer's "selected DU cohorts": 78 men, 55 with BCI below 100, with no significant difference in postoperative residual by DU status; it excluded early revisions, and BCI 100 is the study definition, not a cutoff.
- **AUA24** — AUA/GURS/SUFU incontinence after prostate treatment, amended 2024 (US guideline). Supports further investigation of poor emptying before surgery and counselling about retention risk rather than a numeric residual threshold.
- **ICSUDS17** — ICS good urodynamic practices 2017 (Rosier et al.) (Testing standard). The reason a "technically poor study should not define his future": it sets the quality standards a pressure-flow trace must meet before DU is labelled.
- **CCIFU** — Rigicon ContiClassic instructions for use, REV.03, 2023 (Manufacturer labelling). Requires overflow incontinence to be treated and resolved, and warns about the deactivation and catheterisation plan: the "reliable emptying plan" in the question.

Next → If his rescue plan is a catheter, we have to be clear about what catheterising through a cuff really involves.

13 Catheterisation with a cuff in place

19:00 – 20:30 · 1:30 · The bladder

This slide wants: Make deflation and deactivation the rule before any instrument passes the cuff.

On the slide: The non-negotiable rule for any instrument through a cuff, and the planning around it.

SAY

Intermittent catheterisation does not automatically exclude an AUS, but it changes both the consent and the operation. A small, selected retrospective series supports the feasibility of bulbar AUS in some men who continue catheterisation. It does not demonstrate equivalent safety to men who never catheterise.

Clarify whether the problem is weak emptying, recurrent narrowing or an unresolved outlet that would be better treated first. Establish actual catheter competence, including access and recognition of resistance. We do not want to discover after implantation that the patient can squeeze a pump but cannot carry out the drainage plan.

Any catheterisation requires the cuff to be fully deflated and the device deactivated according to its instructions. Use the least traumatic appropriate instrument, and stop when resistance is unexplained. A deflated cuff's advertised internal diameter is not a safe catheter-size chart, because urethral tissue occupies that space. A 2026 survey of urologists found real gaps in confidence about exactly this step.

Repeated instrumentation and prolonged indwelling drainage need a specialist plan, which may include an alternative drainage route. Observational evidence links prolonged urethral catheterisation with erosion. It does not establish a universally safe duration, or prove that careful intermittent catheterisation carries the same risk.

EXAMPLE

A ward doctor is asked to catheterise a man in retention on the night of day two. The notes say "AUS in situ, deactivated, cuff open" with the surgeon's mobile number. The doctor rings, confirms the device state, uses a small catheter and stops at any resistance. The handover sentence on slide 31 was written for exactly this moment.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **CIC23** — Krughoff et al. 2023: intermittent catheterisation after bulbar AUS (Retrospective selected cohort, 57 men with CIC history). Supports feasibility: 57 men with a CIC history, 18 continuing or starting CIC after AUS, with no significant excess of erosion or removal. Small and selected, so not equivalence.
- **CATH13** — Seideman et al. 2013: prolonged catheterisation and erosion (Observational implantation and revision cohort, 200 men). The reason prolonged access needs a plan: urethral catheterisation over 48 hours was associated with erosion (17 of 44 exposed men). 48 hours is not a guaranteed safe limit.
- **CCIFU** — Rigicon ContiClassic instructions for use, REV.03, 2023 (Manufacturer labelling). The statement on the slide in product form: cuff deflated and device deactivated before any instrumentation, and no catheter-size rule beyond the device instructions.
- **ICS16** — ICS AUS consensus 2015, published 2016 (Biardeau et al.) (International consensus, centred on AMS 800). Device-aware instrumentation, least traumatic instrument, stop on unexplained resistance; AMS-era advice that still frames the principle.
- **BAUSSCI26** — BAUS 2026 abstract P12-7: bulbar AUS in spinal cord injury (Small retrospective conference series). A small 2026 series of bulbar AUS in spinal cord injury, where catheterisation is routine: a specialist discussion point that keeps the drainage route central.
- **INSTR26** — AUS instrumentation knowledge survey 2026 (Girgin, Neurourol Urodyn) (Self-selected survey of 286 Turkish urologists). The footer: a 2026 survey of 286 urologists found confidence gaps in deactivation and instrumentation, which is why competence is tested and a device card is issued. Its catheter sizes are not universal rules.

Next → One more bladder problem remains, and it is the one that can look like success. A dry patient can still be unsafe.

14 Detrusor overactivity and poor compliance

20:30 – 22:00 · 1:30 · The bladder

This slide wants: Tell a transient contraction from a sustained pressure rise, and treat them differently.

On the slide: Two traces, two different problems: a contraction versus a sustained pressure rise.

SAY

Poor compliance needs its own decision. It describes the pressure-volume relationship during filling, not simply the presence of an involuntary contraction. The left trace is a transient contraction; the right is a sustained rise. An apparently dry patient may be storing urine at an unacceptable pressure.

Check the trace and the circumstances. Was there a sustained rise in detrusor pressure, adequate filling and valid subtraction? Consider radiation, fibrosis, neurological disease and the effect of chronic leakage. Assess renal function and upper tracts when there is concern about high-pressure storage or impaired emptying.

Optimise the bladder and repeat the assessment that will show whether storage is acceptable. Refractory cases may need reconstructive or diversion advice before anyone increases outlet resistance. Lai's 2009 cohort made exactly this point: pad results can look good while poor compliance still threatens the upper tracts.

Our four decisions are now clearer. Implant when the leakage mechanism, storage, emptying and device suitability align. Optimise when a treatable bladder problem is dominant. Investigate when the measurements are unreliable or the mechanism uncertain. Reconsider AUS when safe storage, emptying or device operation cannot be achieved. Those categories are a clinical framework and have not been validated as a score.

EXAMPLE

Two men are both dry after an AUS. One had a transient detrusor contraction at 250 mL during filling. The other had a steady pressure rise to 40 cmH2O by 300 mL with no contraction. The first needs urgency treatment if symptomatic. The second needs an upper-tract check and a compliance plan before anyone congratulates him on being dry.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **UDS09** — Lai et al. 2009: urodynamics before AUS (Selected cohort, 129 first AUS after prostatectomy, mean follow-up 34 months). The core warning: in 129 men, adverse preoperative urodynamics did not worsen pad outcomes, but poor compliance could still expose the upper tracts. Dry is not the same as safe.
- **ICS16** — ICS AUS consensus 2015, published 2016 (Biardeau et al.) (International consensus, centred on AMS 800). Recommends evaluating storage pressure and compliance before increasing outlet resistance, and upper-tract assessment where there is concern.
- **BAUS25** — BAUS consensus 2025 (Bhatt et al., BJU Int) (UK expert consensus after literature review and modified Delphi). Its video-urodynamics pathway exists partly to identify poor compliance and unsafe storage before continence surgery.
- **CCIFU** — Rigicon ContiClassic instructions for use, REV.03, 2023 (Manufacturer labelling). Warns that poor bladder compliance is a precaution for the device, so compliance must be addressed before implantation.
- **BAUSSCI26** — BAUS 2026 abstract P12-7: bulbar AUS in spinal cord injury (Small retrospective conference series). A neurogenic (spinal cord injury) series where safe storage was the central prerequisite: the extreme case of the right-hand trace.

Next → That closes the bladder. The urethra is next, and the first job there is simply to name the narrowing correctly.

The urethra

22:00 – 29:00 · 7 minutes · teaching slides 15 to 19

Section slide, no clock time. Read the title, then say: "Name the narrowing, prove the outlet is stable, and plan the tissue the cuff will need."

15 Three different sites of narrowing

22:00 – 23:15 · 1:15 · The urethra

This slide wants: Name the narrowing anatomically: anterior stricture, bladder-neck contracture or VUAS.

On the slide: Three anatomically different narrowings that the word "stricture" hides.

SAY

The shorthand stricture hides different operations and different risks. An anterior urethral stricture involves the urethra and spongiosum. Bladder-neck contracture commonly follows surgery with part of the prostate still present. After radical prostatectomy, the relevant narrowing is usually the vesicourethral anastomosis. Use the anatomy in the discussion and in the referral.

Review every dilation, urethrotomy and reconstruction, including the operative reports. Establish how quickly the narrowing recurred and whether patency currently depends on repeated calibration. Add imaging when the anatomy, length or reconstructive plan requires it.

We call an outlet stable after a credible interval without recurrence, with symptoms, flow, residual and endoscopic findings that support a durable outlet. A scope that happened to pass once does not show that.

Finally, think ahead. A patient who is likely to need further incision, tumour surveillance or treatment for radiation bleeding needs an access plan before a cuff makes those procedures more complicated.

EXAMPLE

A referral says "stricture, dilated twice". On questioning, the man had a radical prostatectomy and the narrowing is at the anastomosis, so it is a VUAS, not an anterior stricture. The reconstructive options, the risk to the cuff site and the guideline pathway all change with that one correction.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **EAUSTR26** — EAU urethral strictures guideline 2026 (European guideline). Provides the three anatomical categories on the slide and the reconstruction pathway for each, including imaging to define site and length.
- **BAUS25** — BAUS consensus 2025 (Bhatt et al., BJU Int) (UK expert consensus after literature review and modified Delphi). Requires narrowing to be treated before continence surgery and a stable outlet to be shown, which starts with naming the narrowing.
- **AUA24** — AUA/GURS/SUFU incontinence after prostate treatment, amended 2024 (US guideline). Recommends treating BNC or VUAS before SUI surgery with interval cystoscopy afterwards: the plan

depends on the correct name.

Next → Stability is a finding, not a date on the calendar. Here is what the three guidelines actually say about the interval.

16 Reassessment and urethral stability

23:15 – 24:45 · 1:30 · The urethra

This slide wants: Call the outlet stable only on findings, and match each guideline interval to its own pathway.

On the slide: Three intervals from three guidelines, each describing a different pathway.

SAY

BAUS recommends waiting at least six months after treating a post-prostatectomy narrowing, with no recurrence, before continence surgery. That is a genuine consensus recommendation, although it is not a randomised demonstration that six months is the unique safe interval.

The AUA discussion describes interval cystoscopy at least four to six weeks after treatment of symptomatic bladder-neck contracture or anastomotic stenosis. That is a point for reassessment, with no guarantee that every reconstructed or irradiated urethra is ready for a cuff. EAU's three-to-six-month advice after redo vesicourethral anastomosis belongs to that particular reconstructive setting.

Staged treatment lets us observe recurrence and avoid instrumenting a newly implanted cuff. Synchronous treatment has been reported in selected expert series, but should not become the default for aggressive, recurrent disease.

Our decision would use the recurrence pattern, tissue quality, planned future access and the guideline context. A patient who has repeatedly restenosed after a short interval has not become low risk because the calendar reaches a convenient date.

EXAMPLE

A man had a VUAS incised in January and a clear cystoscopy in March (the AUA's four-to-six-week reassessment). Under BAUS the earliest implant would be July, six months without recurrence. If he restenosed in May, the calendar resets, and the pattern of early recurrence matters more than any date.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **BAUS25** — BAUS consensus 2025 (Bhatt et al., BJU Int) (UK expert consensus after literature review and modified Delphi). The BAUS row: at least six months without recurrence after treatment of post-prostatectomy narrowing, as consensus.
- **AUA24** — AUA/GURS/SUFU incontinence after prostate treatment, amended 2024 (US guideline). The AUA row: interval cystoscopy at least four to six weeks after symptomatic BNC or VUAS treatment, as a reassessment point.
- **EAUSTR26** — EAU urethral strictures guideline 2026 (European guideline). The EAU row: three to six months after redo vesicourethral anastomosis, which belongs to that reconstruction, not to every dilation.
- **ICS16** — ICS AUS consensus 2015, published 2016 (Biardeau et al.) (International consensus, centred on AMS 800). Advises demonstrated outlet stability and staged treatment to avoid instrumenting a new cuff:

the footer's "recurrence pattern, tissue quality and future access".

Next → Our third patient brings all of that together, with radiation added.

17 Case 3: radiation and recurrent VUAS

24:45 – 26:15 · 1:30 · The urethra

This slide wants: Choose between implanting, staging reconstruction and reconsidering the endpoint after radiation.

On the slide: Case 3: an irradiated, repeatedly re-stenosing anastomosis and a man who wants to be dry now.

SAY

This fictional patient has had radical prostatectomy, salvage radiotherapy and repeated treatment of an anastomotic stenosis. He leaks severely when the outlet is open, but the narrowing returns. He understandably wants continence now.

Would you put an AUS into that pathway? [Pause for six seconds.] Our first answer is to defer the implant and obtain a specialist reconstructive plan. Current patency is not durable, and more endoscopic treatment is likely. We also need to understand the bladder, including compliance and the burden of radiation cystitis.

Radiation is associated with worse continence and greater erosion or explantation risk in pooled observational AUS studies, roughly a doubling of the odds of erosion. Those estimates combine different follow-up periods and case mixes, so they cannot give his personal probability of failure. A 2025 cohort found the same association with cuff removal and hinted at a timing effect, but that trend was not significant and sets no waiting rule.

Definitive reconstruction followed by demonstrated stability and later AUS may be reasonable. In a devastated outlet with unsafe or intolerable bladder function, a different drainage or diversion strategy may ultimately offer the better life. There is more than one defensible endpoint. Repeatedly reopening a narrow lumen is not necessarily progress if it stops us agreeing a durable plan.

EXAMPLE

Case 3 is on the slide. A good answer sounds like this: "Stage. Current patency depends on the next incision, so I would not put a cuff on it. Reconstruct or at least stabilise the anastomosis, check compliance and radiation cystitis, then implant when stability is demonstrated. If the bladder is unsafe, a diversion conversation is a legitimate endpoint." Radiation roughly doubles the odds of erosion in pooled data; that goes into consent, not into a refusal.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **RT22** — Zhang and Xu 2022: radiation meta-analysis (Meta-analysis of 18 observational studies). The footer's "radiation increases risk": pooled erosion or explantation odds about 2.0 (95% CI 1.15–3.45) in 18 observational studies. An odds ratio, not his personal probability.
- **BAUS25** — BAUS consensus 2025 (Bhatt et al., BJU Int) (UK expert consensus after literature review and modified Delphi). Requires a stable outlet and treatment of narrowing before continence surgery, which is why "stage" is the first answer.

- **EAUSTR26** — EAU urethral strictures guideline 2026 (European guideline). Gives the reconstructive pathway for recurrent VUAS, including the interval after redo anastomosis, and frames diversion as an endpoint when the outlet is devastated.
- **AUA24** — AUA/GURS/SUFU incontinence after prostate treatment, amended 2024 (US guideline). Favours AUS over sling in irradiated men and asks for interval cystoscopy after VUAS treatment; counselling about higher erosion risk is explicit.
- **RTTIME25** — Radiation and cuff-removal timing cohort 2025 (Lee et al., World J Urol) (Retrospective cohort, 283 men). The footer's "2025 timing study": radiation was associated with cuff removal (HR 3.30) in 283 men, and the trend towards later implantation was not significant, so no waiting rule follows.

Next → Suppose he does get his reconstruction. What does a repaired urethra mean for the operation we would do later?

18 AUS after urethral reconstruction

26:15 – 27:45 · 1:30 · The urethra

This slide wants: Read the urethroplasty history and plan the dissection around the tissue that is left.

On the slide: How a previous urethroplasty changes the tissue available and the dissection we plan.

SAY

Previous urethroplasty covers several different operations. Was the urethra transected? Was there a graft, a posterior reconstruction, a prior cuff erosion or a fistula? Where is the viable spongiosum now? Those details should change where and how we dissect.

The operation must preserve what remains. Avoid extending mobilisation through a repaired segment unless it is needed. A technically convenient cuff site may have poor tissue. If a safe plane cannot be established, pause and reconsider the approach or stage the operation.

Davis's multicentre cohort reports very different later AUS outcomes after transecting and non-transecting urethroplasty, and the 2026 meta-analysis of 533 men points the same way: prior urethroplasty roughly doubles explantation, and transecting repairs carry more erosion and explantation. It adds evidence beyond the older guidance, but it remains observational. Stricture severity, anatomy, selection and unequal follow-up confound the comparison. It supports a more detailed reconstructive history and careful counselling; it cannot support a claim that every transection was avoidable.

The practical point is to plan the reconstruction and any future continence surgery together. After previous erosion and explantation, healing, patency and realistic reimplantation expectations need to be demonstrated again.

EXAMPLE

Before an AUS, the operative note of a bulbar urethroplasty three years earlier is obtained. It was an excision and primary anastomosis, so the urethra was transected. The plan changes: the cuff site is moved away from the anastomosis, the dissection is kept to the minimum, and the patient is told that explantation was reported in 56% after transecting repairs in the Davis series, against 24% after non-transecting ones.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **URETH26** — Davis et al. 2026: AUS after urethroplasty (Retrospective multicentre series, 178 cases, 15 institutions). The multicentre figure: explantation-level complications in 56.2% after transecting versus 23.5% after non-transecting urethroplasty in 178 cases. Confounded by severity and selection; it supports a detailed history, not blame.
- **BAUS25** — BAUS consensus 2025 (Bhatt et al., BJU Int) (UK expert consensus after literature review and modified Delphi). Asks for the reconstruction history and a stable outlet before an AUS, and specialist planning after previous erosion.
- **AUA24** — AUA/GURS/SUFU incontinence after prostate treatment, amended 2024 (US guideline). Supports counselling about higher erosion risk after urethral reconstruction and planning the cuff site accordingly.
- **ICS16** — ICS AUS consensus 2015, published 2016 (Biardeau et al.) (International consensus, centred on AMS 800). Operative principle of preserving spongiosum and attachments and choosing viable cuff-bearing tissue: the right-hand column.
- **URETHMETA26** — Prior urethroplasty meta-analysis 2026 (Deameh et al., Urology) (Meta-analysis of four observational studies, 533 men). The footer: in 533 men, prior urethroplasty roughly doubled explantation (RR 2.05) and transecting repairs carried more erosion (RR 2.34) and explantation (RR 2.38). Observational and overlapping with Davis.

Next → When the dorsal plane is hostile, one option is to go through the corpora. It is an option with a price.

19 Transcorporal placement: role and costs

27:45 – 29:00 · 1:15 · The urethra

This slide wants: Use transcorporal placement for a defined dissection problem, and name its price.

On the slide: Transcorporal placement: a manoeuvre for a defined problem, with real costs and uncertain protection.

SAY

Transcorporal placement can avoid a hazardous dorsal dissection and include tunical tissue beneath part of the cuff. That is a plausible reconstructive purpose, but it does not prove protection from erosion.

The comparative retrospective evidence points in different directions. Redmond's fragile-urethra series favoured it numerically. Kurtzman's high-risk series found more infection or erosion-related explantation in the transcorporal group. The latest systematic review pools twenty studies and suggests less erosion but more revision. Surgeons tend to select it for the worst urethras, so confounding by indication is substantial in both directions.

Discuss corporal bleeding, erectile consequences and the implications for an existing or future penile prosthesis. Sizing must account for the tissue actually enclosed, using the relevant device protocol and no improvised correction factor.

Tandem cuffs, wraps, relocation and capsular procedures also consume tissue or add complexity. Choose a manoeuvre because it addresses a demonstrated problem, and state its limitation. If the proposal is simply to squeeze a failing urethra harder, the diagnosis and

the endpoint need another look.

EXAMPLE

During a revision the dorsal plane behind a previously eroded segment will not open safely. The surgeon stops, and either changes level or goes transcorporal with the patient's prior consent to the sexual and bleeding consequences. What the surgeon does not do is say "transcorporal protects against erosion", because the comparative data point both ways.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **TC20** — Redmond et al. 2020: transcorporal placement in fragile urethras (Retrospective series, 76 patients, 24 transcorporal). The series that favours the manoeuvre numerically: revision 20.8% versus 36.5% and erosion 8.3% versus 17.3% in fragile urethras, with the erosion difference not significant and no randomisation.
- **TC23** — Kurtzman et al. 2023: high-risk transcorporal AUS (Retrospective single-surgeon high-risk cohort, 68 men, 77 cuffs). The series that points the other way: more infection or erosion explants after transcorporal placement (HR 2.74) in a high-risk cohort, probably because the worst urethras were selected for it.
- **BAUS25** — BAUS consensus 2025 (Bhatt et al., BJU Int) (UK expert consensus after literature review and modified Delphi). Recognises transcorporal placement as an option for a compromised urethra in specialist hands, without a comparative claim.
- **AP23** — Asia-Pacific AMS 800 consensus 2023 (Chung et al.) (Regional expert consensus). Describes transcorporal and tandem-cuff techniques as salvage options with sizing that must follow the device protocol, and lists their sexual and corporal consequences.
- **TCSYS25** — Transcorporal placement systematic review, online 2025 (Domínguez Gutiérrez et al.) (Systematic review, 20 studies, eight comparative). The footer: 20 studies pooled gave less erosion (OR 0.35) and infection but more revision (OR 2.99), with no explantation difference. Non-randomised and heterogeneous, so no causal protection.

Next → That is the urethra. Now a different kind of decision: the things we do before the skin is opened, and whether the evidence supports them.

Preparation without ritual

29:00 – 36:00 · 7 minutes · teaching slides 20 to 24

Section slide, no clock time. Read the title, then say: "Identify the activity, the product and the evidence before debating a duration."

20 The fifteen-minute wash question

29:00 – 30:15 · 1:15 · Preparation without ritual

This slide wants: Say which preparation activity you mean before debating how long it takes.

On the slide: The fifteen-minute wash: first identify which activity we mean.

SAY

Is a fifteen-minute wash or scrub necessary before AUS implantation? We found no direct evidence that establishes a universal requirement.

NICE advises soap washing the day before or the day of surgery. That is different from operative skin preparation, surgical hand antisepsis, wound irrigation, external device dipping and internal device filling. Identify the activity before debating its duration.

BAUS recommends a five-minute topical field scrub by Delphi consensus, and acknowledges that this is extrapolated from penile prosthesis and other implant surgery. It does not establish fifteen minutes as an AUS standard. For hands, use the selected product's surgical antisepsis instructions. For the patient's skin, use the appropriate product and its application, contact and complete drying requirements.

Waiting fifteen minutes cannot compensate for missed skin, an unsuitable product, pooled alcohol or a contaminated field. The lack of evidence for this ritual does not make antisepsis dispensable. We need correct, reproducible preparation.

EXAMPLE

A scrub nurse asks whether the fifteen-minute wash has been done. Reply by asking which activity: the patient's shower this morning (NICE), the alcohol-chlorhexidine skin preparation that needs full drying (product information), the team's hand antisepsis (WHO) or the device dip (manufacturer protocol). Each has its own instruction. None is a fifteen-minute AUS rule.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **BAUS25** — BAUS consensus 2025 (Bhatt et al., BJU Int) (UK expert consensus after literature review and modified Delphi). The footer: a five-minute topical field scrub by Delphi consensus, openly extrapolated from other implant surgery. It is the only AUS-specific duration in guidance and it is not fifteen minutes.
- **NICE** — NICE NG125: surgical site infection (UK national guidance). Row one (patient washing with soap before surgery) and row two (antiseptic skin preparation with drying); it tests no fifteen-minute AUS routine.
- **WHO09** — WHO hand hygiene guidance 2009 (International standard). Row three: surgical hand antisepsis is a separate activity with its own product instructions.
- **PREP24** — Bourgi et al. 2024: AMS 800 skin-preparation protocols (Retrospective sequential-era comparison, 156 procedures). The only AUS preparation comparison: intensified povidone preparation

showed no association with device removal (OR 0.97, wide CI), and fifteen minutes was never tested against shorter.

- **IRRIG26** — Gentamicin withdrawal cohort 2026 (Skrzypczyk et al., Antibiotics) (Single-centre before-after study, 173 uncoated AMS 800 procedures). Row four: irrigation and device dipping are separate interventions; the gentamicin-withdrawal cohort (1 of 59 versus 2 of 114 infections) cannot prove anything about either.

Next → What the studies can actually support is narrower than the folklore, and here is the best of them.

21 Skin culture reduction and clinical infection

30:15 – 32:00 · 1:45 · Preparation without ritual

This slide wants: Separate a skin-culture result from a clinical-infection result.

On the slide: One good randomised trial with a surrogate endpoint: skin cultures, not implant infections.

SAY

In a randomised study of one hundred patients having initial genitourinary prosthetic surgery, chlorhexidine-alcohol produced fewer positive post-preparation skin cultures than povidone-iodine. The reported proportions were eight versus thirty-two per cent. That is a microbiological endpoint in a mixed prosthesis population. The study did not show fewer ContiClassic infections.

A later retrospective AMS 800 study compared different povidone preparation protocols. It did not show an advantage for the intensified protocol, but era effects, reporting inconsistencies and limited follow-up reporting weaken the inference. It did not compare fifteen minutes against a shorter preparation.

General SSI guidance supports an appropriate alcohol-based preparation where suitable, with alternatives for contraindications or anatomical requirements. The exact product matters around genital skin and mucosa. Check allergy, permitted application sites and the product information before applying a study's formulation to every field.

Do not remove hair routinely. If it is required for exposure or adhesion, clipping on the day is the general NICE recommendation, and avoid skin trauma. Before draping and diathermy, confirm drying and the absence of pooling in skin folds or under the patient. Those are practical safety requirements, whatever the stopwatch says.

EXAMPLE

A colleague quotes "chlorhexidine cuts AUS infection four-fold". The actual finding is 8% versus 32% positive skin cultures after preparation, in 100 mixed genitourinary prosthetic procedures. That supports choosing an alcohol-chlorhexidine preparation where the product permits it. It does not show fewer AUS infections and tested no duration.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **YEUNG13** — Yeung et al. 2013: GU prosthetic skin preparation RCT (Single-centre randomised trial, 100 initial GU prosthetic procedures, 50 per arm). The chart: 8% versus 32% positive post-preparation skin cultures with chlorhexidine-alcohol versus povidone-iodine in 100 initial GU prosthetic cases. A surrogate endpoint; no AUS infection difference and no duration tested.
- **PREP24** — Bourgi et al. 2024: AMS 800 skin-preparation protocols (Retrospective sequential-era comparison, 156 procedures). The second study in the narration: a sequential-era povidone comparison with no advantage for the intensified protocol and wide uncertainty.
- **NICE** — NICE NG125: surgical site infection (UK national guidance). Supports an appropriate alcohol-based antiseptic where suitable, clipping rather than shaving, and complete drying before diathermy.
- **CHLORA26** — ChloraPrep Swabstick UK product information, 2026 (Product information (SmPC)). Why the exact product matters: a 30-second application, intact skin only, permitted sites and flammability warnings for that presentation. Not a universal genital preparation prescription.
- **BAUS25** — BAUS consensus 2025 (Bhatt et al., BJU Int) (UK expert consensus after literature review and modified Delphi). Its five-minute scrub recommendation sits alongside this evidence as consensus, which is why the slide separates culture reduction from clinical infection.

Next → Skin is one disagreement. Urine and antibiotics are another, and here the sources genuinely conflict.

22 Urine and antibiotic disagreements

32:00 – 33:30 · 1:30 · Preparation without ritual

This slide wants: State the urine and antibiotic disagreements openly and agree a local written policy.

On the slide: Four questions where guidance, evidence and the device label disagree, stated openly.

SAY

Treat symptomatic urinary infection and reassess before elective implantation. Asymptomatic bacteriuria is different.

BAUS asks for a preoperative urine sample and treatment of infection. IDSA specifically suggests against screening for or treating asymptomatic bacteriuria solely for AUS implantation, although that recommendation rests on very low-quality evidence. Procedures involving urinary mucosal trauma are a separate situation.

Agree the urine pathway with local microbiology, and do not equate every positive culture with symptomatic infection. Choose prophylaxis using the implant policy, previous isolates, allergies and renal function. Complete administration at the appropriate time before incision, and redose when the agent and procedure require it.

Observational AUS evidence has not shown a convincing benefit from routine postoperative oral antibiotics; Adamsky's claims cohort found explantation rates of 3.9 and 4.0 per cent with and without them. However, the retrieved ContiClassic IFU describes forty-eight hours of intravenous antibiotics and five days of oral antibiotics after discharge as general care. That conflicts with shorter-prophylaxis stewardship approaches. We would respond by verifying current regional labelling and agreeing a documented protocol with the implant service and microbiology. Do not silently ignore either source, and do not describe an

extended regimen as universally proven to prevent infection.

EXAMPLE

The pre-admission urine culture grows 10^5 E. coli; the patient has no symptoms. BAUS would treat before implantation; IDSA would not screen or treat for an AUS alone. The agreed local policy, written with microbiology, decides, and the decision is documented either way. The same written policy says what to do with the IFU's five-day oral course after discharge.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **IDSA19** — IDSA asymptomatic bacteriuria guideline 2019 (Nicolle et al.) (US infectious-disease guideline). Row two: suggests against screening for or treating asymptomatic bacteriuria solely for an AUS, on very low-quality evidence, which differs from the BAUS urine pathway.
- **BAUS25** — BAUS consensus 2025 (Bhatt et al., BJU Int) (UK expert consensus after literature review and modified Delphi). Row one and two: a preoperative urine sample with treatment of infection before implantation.
- **AUAABX20** — AUA antimicrobial prophylaxis statement 2020 (Lightner et al.) (Consensus statement informed by literature review). Row three: prophylaxis completed before incision, redosed when needed, and no unnecessary continuation.
- **ABX18** — Adamsky et al. 2018: postoperative oral antibiotics (Claims-based observational study, 3,594 AUS recipients). Row four's "older observational data": explantation 3.9% with and 4.0% without postoperative oral antibiotics in 3,594 recipients, with no protective association.
- **CCIFU** — Rigicon ContiClassic instructions for use, REV.03, 2023 (Manufacturer labelling). Row four's "retrieved IFU differs": 48 hours of intravenous and five days of oral antibiotics after discharge, which conflicts with stewardship sources and must be reconciled locally.
- **AUAABX26** — AUA 2026 abstract IP09-18: postoperative antibiotics (Conference abstract, title and programme verified only). The footer: a 2026 abstract whose title states no infection benefit from postoperative antibiotics; results unverified, nothing numerical inferred.

Next → Once the patient is prepared, the remaining risk is what we do with our hands and the implant.

23 Contamination control during implantation

33:30 – 35:00 · 1:30 · Preparation without ritual

This slide wants: Reduce contamination with four practical controls, offered as rationale, not proof.

On the slide: Four practical contamination controls, offered as rationale rather than proof.

SAY

There are sensible ways to reduce contamination. Prepare the equipment before opening the implant, establish a protected sterile handling area, minimise skin contact, and avoid unnecessary traffic and repeated hand-offs.

Our preference is fresh sterile gloves before handling the device and after any contamination concern. We offer that as a practical control, without claiming that each glove change has been proven to reduce AUS infection. Minimal-touch AUS series are encouraging, but they compare different eras and unequal follow-up. We cannot attribute their results to one drape or one manoeuvre.

Separate wound irrigation from device dipping and internal filling. General NICE guidance does not support wound irrigation solely to prevent SSI. AUS consensus literature also acknowledges uncertainty about antibiotic irrigants. The 2026 gentamicin-withdrawal cohort saw one infection in 59 procedures with gentamicin and two in 114 without; three events cannot prove equivalence, and immersion and irrigation changed together. A hydrophilic coating's ability to absorb a solution does not establish that any chosen antibiotic mixture improves clinical outcomes.

For ContiClassic, follow the verified device instructions and obtain the operative protocol for detailed preparation. Do not substitute a remembered AMS 800 recipe or improvise an unverified antiseptic bath for an implant.

EXAMPLE

The implant stays in its packaging until the field is ready. The surgeon changes gloves before touching it, the device is handled on a separate covered tray, and the wound irrigant is kept apart from whatever the device is dipped in. If asked which of those four steps prevents infection, the honest answer is "the bundle has rationale; no single step is proven".

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **TOUCH23** — Ziegelmann et al. 2023: minimal-touch AUS technique (Historical comparison of a contamination-control bundle). The bundle behind steps one to three: crude infections fell after a minimal-touch technique, but twelve-month differences were not significant, so no single glove change or pause can be credited.
- **NICE** — NICE NG125: surgical site infection (UK national guidance). Step four: no wound irrigation solely to reduce SSI, which is why irrigation is separated from device dipping.
- **AP23** — Asia-Pacific AMS 800 consensus 2023 (Chung et al.) (Regional expert consensus). Describes the AMS 800 preparation and antibiotic-soak sequence and acknowledges uncertainty about irrigants; its recipe does not transfer to ContiClassic.
- **CCIFU** — Rigicon ContiClassic instructions for use, REV.03, 2023 (Manufacturer labelling). Directs detailed preparation to the separate OR protocol and forbids improvised handling, which is why a remembered AMS 800 recipe is not used.
- **IRRIG26** — Gentamicin withdrawal cohort 2026 (Skrzypczyk et al., Antibiotics) (Single-centre before-after study, 173 uncoated AMS 800 procedures). The footer: 90-day SSI 1 of 59 with gentamicin versus 2 of 114 without, in uncoated AMS 800 procedures; three events, immersion and irrigation changed together, so no equivalence and no transfer to ContiClassic.

Next → All of this collapses into one checklist that the whole team runs before the implant is opened.

24 The pre-implant theatre checklist

35:00 – 36:00 · 1:00 · Preparation without ritual

This slide wants: Run one pre-implant checklist and know which rows are guidance and which are unresolved.

On the slide: The checklist we would actually use, with each row labelled by the kind of authority behind it.

SAY

Before opening the implant, confirm the patient, intended operation, tissue concerns and stop conditions. Review urinary and skin infection concerns, prophylaxis delivery and the redosing plan. Confirm that the preparation is anatomically suitable and dry.

Then confirm the exact device and compatible components, available cuff sizes, planned pressure range, current instructions and the person responsible for preparation. Agree the catheter and postoperative rescue plan.

Finally, establish what happens if the urethra is injured or sterility is compromised. A checklist is most useful when it makes that response predictable. Notice the last row: a fixed fifteen minutes, the best irrigant and prolonged prophylaxis sit there as unresolved evidence, not as requirements.

Audit completion of these steps. The duration of a wash is not a measure of the quality of the whole operation.

EXAMPLE

Before the implant is opened the team runs the five rows aloud: indication and infection review (guidance), antiseptic and antibiotic choice (product and local policy), model, sizes and pressure (device instructions), what happens on contamination (expert implementation), and the fifteen-minute wash (unresolved). The last row is read out precisely so nobody mistakes it for a requirement.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **NICE** — NICE NG125: surgical site infection (UK national guidance). The "guidance-supported" basis for infection review, asepsis, antiseptic choice and drying.
- **BAUS25** — BAUS consensus 2025 (Bhatt et al., BJU Int) (UK expert consensus after literature review and modified Delphi). The "guidance-supported" basis for indication review and prophylaxis, and the source of the five-minute scrub that sits in the unresolved row.
- **CCIFU** — Rigicon ContiClassic instructions for use, REV.03, 2023 (Manufacturer labelling). The "current device instructions" row: model, sizes, pressure and preparation must follow the ContiClassic label and its OR protocol.
- **AUAABX20** — AUA antimicrobial prophylaxis statement 2020 (Lightner et al.) (Consensus statement informed by literature review). The "product and local policy" row: antibiotic choice, timing and redosing.
- **IRRIG26** — Gentamicin withdrawal cohort 2026 (Skrzypczyk et al., Antibiotics) (Single-centre before-after study, 173 uncoated AMS 800 procedures). The "unresolved evidence" row: the best irrigant is not settled by a three-event cohort.

Next → Now into theatre. The next eight slides follow the operation in order, and the model will show us each space.

The operation

36:00 – 50:00 · 14 minutes · teaching slides 25 to 32

Section slide, no clock time. Read the title, then say: "Preserve tissue, measure what the cuff will enclose, follow the actual device protocol, and verify before closing."

25 Positioning and exposure

36:00 – 37:15 · 1:15 · The operation

This slide wants: Position and expose for the anatomy in front of you, planning all three component sites together.

On the slide: Positioning that serves the anatomy we have, and an access plan for all three components.

SAY

Positioning should expose the bulbar urethra while protecting pressure points and respecting hip mobility. Check the patient before anaesthesia when restricted movement is likely to matter. Do not accept excessive flexion simply because it improves the first view.

For a standard male bulbar implant, a single-cuff perineal approach has guideline support. Other approaches may be appropriate for selected anatomy and experienced operators, but convenience alone should not determine access to a compromised urethra.

Plan where the balloon and pump will go at the same time. Previous pelvic surgery, inguinal mesh, abdominal scars and an existing penile prosthesis alter that plan. Have the relevant operative records available.

If exposure is poor, improve the incision, retraction or positioning deliberately. Excessive traction to compensate for a small working space transfers force to the urethra. The purpose of exposure is to make the next decision visible, and a shorter incision is not worth losing that.

EXAMPLE

A man with a stiff hip and inguinal mesh on the left is examined before anaesthesia. Lithotomy is set to what his hips tolerate, the balloon is planned away from the mesh, and the pump side is the right, which he chose in clinic. All three sites are decided before the perineal incision.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **AUA24** — AUA/GURS/SUFU incontinence after prostate treatment, amended 2024 (US guideline). Supports a single-cuff perineal bulbar approach as the standard primary operation, with alternatives for selected anatomy.
- **ICS16** — ICS AUS consensus 2015, published 2016 (Biardeau et al.) (International consensus, centred on AMS 800). Operative principles: positioning to expose the bulbar urethra, protecting pressure points and planning the balloon and pump sites in advance.
- **AP23** — Asia-Pacific AMS 800 consensus 2023 (Chung et al.) (Regional expert consensus). Practical consensus on exposure, previous surgery, mesh and an existing penile prosthesis altering the access plan.

Next → With the urethra exposed, the next decisions are about which tissue the cuff will sit on, and how much of it we disturb.

26 Bulbar dissection and tissue preservation

37:15 – 38:45 · 1:30 · The operation

This slide wants: Expose a viable cuff-bearing segment and mobilise no more than the cuff needs.

On the slide: The cuff-bearing segment: enough controlled exposure, and no more mobilisation than the cuff needs.

SAY

The goal is a cuff around a viable urethral segment, not a completely skeletonised urethra. In a straightforward case, the proximal bulbar region often offers useful spongiosal tissue. Prior repair, erosion or scarring may change which segment is safest.

The dorsal relationship to the corporal bodies is where a small error can become a full-thickness injury. Use controlled dissection with a clear view of the plane. Do not pass an instrument blindly behind the urethra and assume that resistance is a harmless adhesion.

Preserve spongiosum and vascular attachments, limit circumferential mobilisation to what the cuff requires, and control bleeding without indiscriminate thermal injury. In scarred tissue, repeated forceful spreading can do more damage than a deliberate change of approach.

If the plane disappears, stop. Improve the view, reassess the site or seek reconstructive help. Transcorporal placement may be an option in selected hands, but it should be a considered plan with consent and appropriate expertise. Tissue preservation has a clear biological rationale. The contribution of each manoeuvre to erosion prevention has not been separated in high-quality comparative trials.

EXAMPLE

The proximal bulbar urethra is exposed and a right-angle clamp meets resistance dorsally. The surgeon does not push. The view is improved, the plane is reassessed under direct vision, and the segment is moved slightly if the plane is scarred. The urethra is never skeletonised beyond the cuff width.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **ICS16** — ICS AUS consensus 2015, published 2016 (Biardeau et al.) (International consensus, centred on AMS 800). The operative principle of a viable cuff-bearing segment with preserved spongiosum and controlled dorsal dissection.
- **AP23** — Asia-Pacific AMS 800 consensus 2023 (Chung et al.) (Regional expert consensus). Describes the proximal bulbar site and the dorsal plane hazards, and when to change level or consider transcorporal placement.
- **CCIFU** — Rigicon ContiClassic instructions for use, REV.03, 2023 (Manufacturer labelling). Warns that thin spongiosum and compromised tissue are precautions for the device: the reason "uncertain integrity prompts reassessment".

- **URETH26** — Davis et al. 2026: AUS after urethroplasty (Retrospective multicentre series, 178 cases, 15 institutions). The reason scarring changes the plan: explantation was far higher after transecting urethroplasty, so a repaired segment is treated as fragile.

Next → Once the segment is chosen, we measure it. And measurement is where the most avoidable errors live.

27 Cuff measurement and fit

38:45 – 40:30 · 1:45 · The operation

This slide wants: Measure the tissue the cuff will actually enclose, with the device's own method.

On the slide: Measuring the tissue the cuff will actually enclose, with the device's own method.

SAY

Measurement fails when the tissue being measured is not the tissue the cuff will finally enclose. Retraction can stretch the urethra, and a tight measuring tape can compress it. Included fat, scar or muscle can enlarge the apparent circumference, and haematoma and tissue distortion add further uncertainty.

Relax the traction, confirm the plane and use the manufacturer's sizing method. Recheck an implausible reading; do not select the familiar size from memory. In a reconstruction or transcorporal operation the enclosed tissue is different, so do not apply an unsupported numerical adjustment simply because it is traditional.

Too small a cuff can cause excessive compression, retention and tissue injury. Too large a cuff can leave poor coaptation and persistent leakage. Neither problem is reliably solved by choosing a stronger pressure-regulating balloon. Assess the final relationship and device function, with an appropriate integrity check when indicated. If the finding and the measurement disagree, investigate the discrepancy.

The early ContiClassic safety series included revisions attributed to sizing errors, four of its eight. That reinforces the importance of this step without telling us which sizing strategy is superior. More available sizes may help a surgeon choose a fit. Their availability alone is not evidence of better long-term continence or less erosion.

EXAMPLE

The tape reads 3.5 cm while an assistant holds the urethra on traction with a loop. Traction is released and the tape is reapplied: 4.0 cm. The second number is the one the cuff will enclose. Four of the eight revisions in the first ContiClassic series were sizing errors, which is why the reading is rechecked rather than trusted.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **ICS16** — ICS AUS consensus 2015, published 2016 (Biardeau et al.) (International consensus, centred on AMS 800). Sizing principle: measure the relaxed urethra without traction and choose the cuff that fits the enclosed tissue.
- **CCIFU** — Rigicon ContiClassic instructions for use, REV.03, 2023 (Manufacturer labelling). Requires the system's own sizing method and available cuff sizes; no adjustment factor is specified for altered tissue.

- **CCSAFETY** — Wilson et al.: first ContiClassic safety outcomes, IJIR 2024 (Uncontrolled early device series, 116 implants, mixed indications). Four of the eight revisions in the first 116 ContiClassic implants were attributed to sizing errors, which is why the reading is rechecked. It does not say which sizing strategy is better.
- **AP23** — Asia-Pacific AMS 800 consensus 2023 (Chung et al.) (Regional expert consensus). Warns against increasing balloon pressure to compensate for a poor cuff fit: the last bullet.

Next → Sometimes the dissection or the measurement reveals something that changes what today's operation is for.

28 Urethral injury or contamination

40:30 – 42:00 · 1:30 · The operation

This slide wants: Change the endpoint of the operation when the urethra is injured or the field is contaminated.

On the slide: The stop conditions: a recognised injury or a lost sterile field changes the endpoint.

SAY

Suspect injury when the plane is wrong, the wall appears breached, fluid escapes or there has been difficult instrumentation. Inspection, an appropriate controlled leak test and cystoscopy can help, but a negative test does not make aggressive dissection harmless. No single test has proven superiority for preventing erosion.

Recognised urethral injury should change the objective of the operation. Stop implantation, define the injury, repair when appropriate and establish suitable drainage. Explain the change to the patient afterwards, and plan reassessment of healing and patency before considering another implant. Consensus intervals are guides. The injury and the reconstruction determine the actual pathway.

For contamination, stop handling the device and identify what has been contaminated: a glove, instrument, component or the operative field. Replace contaminated disposable components and restore a clean setup as appropriate. Gross contamination or an uncertain tissue field may justify abandoning implantation and staging it.

These are single-use implants. A dropped or contaminated component should not return to service after an improvised antiseptic soak. Finishing with no implant can be the correct outcome when the conditions for safe implantation have gone.

EXAMPLE

Fluid appears in the dorsal plane during dissection and cystoscopy shows a urethral breach. The implant is not opened. The injury is repaired, a catheter is left, and the patient is told afterwards that the operation changed. Reimplantation is discussed only after healing and patency are shown again. If instead a cuff had touched an unsterile drape, it would be discarded, not soaked in antiseptic.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **BAUS25** — BAUS consensus 2025 (Bhatt et al., BJU Int) (UK expert consensus after literature review and modified Delphi). Supports stopping implantation and staging after urethral injury, with reassessment of healing before reimplantation.

- **AP23** — Asia-Pacific AMS 800 consensus 2023 (Chung et al.) (Regional expert consensus). Describes leak testing, cystoscopy and the response to contamination, including replacing contaminated components.
- **ICS16** — ICS AUS consensus 2015, published 2016 (Biardeau et al.) (International consensus, centred on AMS 800). Operative principle: recognised injury changes the operation to repair and drainage; no single integrity test is proven to prevent erosion.
- **CCIFU** — Rigicon ContiClassic instructions for use, REV.03, 2023 (Manufacturer labelling). Single-use labelling: a contaminated or dropped component must not be reused after an improvised soak, as the footer says.

Next → If the cuff is safely placed, two separate decisions remain for the balloon: where it goes, and which pressure component we choose.

29 Balloon pocket and pressure selection

42:00 – 43:30 · 1:30 · The operation

This slide wants: Treat the balloon pocket and the pressure component as two separate decisions.

On the slide: Balloon pocket and pressure component: two decisions, two device-specific labels.

SAY

A routine retropubic pocket can become hazardous after pelvic surgery, radiation, mesh repair or previous implants. Review that anatomy before entering the space. Do not overcome unexpected resistance with a stronger finger or instrument: bladder, bowel and vessels may be close to the intended route.

An alternative ectopic or submuscular position may be appropriate in experienced hands, with attention to the device's permitted placement, secure coverage, access and tubing route. These approaches also have limitations, including malposition, palpability and herniation. A different pocket is only safer if it is created correctly.

Pressure selection is a separate issue. ContiClassic labelling lists ranges including sixty to sixty-nine centimetres of water. AMS 800 consensus commonly discusses sixty-one to seventy. Those labels must not be treated as interchangeable operative instructions.

The public ContiClassic IFU confirms sterile saline filling but directs detailed implantation to a separate Operating Room Protocol. Without that protocol we cannot verify a filling volume or a universal pressure-selection algorithm. Confirm both before surgery. And increasing pressure should never be a substitute for diagnosing a poor cuff fit or a failing urethra.

EXAMPLE

A man with a previous open prostatectomy and bilateral hernia mesh needs a balloon pocket. The retropubic route is abandoned for a deliberately created submuscular pocket. Separately, the pressure component is chosen from the ContiClassic label (60–69 cmH2O range on the slide), not from an AMS 800 habit, and the fill volume is taken from the current OR protocol, not from memory.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **CCIFU** — Rigicon ContiClassic instructions for use, REV.03, 2023 (Manufacturer labelling). The left panel: the ContiClassic label's 60–69 cmH2O range, sterile saline filling, and the direction to a separate OR protocol for volumes and selection, which could not be verified.
- **AP23** — Asia-Pacific AMS 800 consensus 2023 (Chung et al.) (Regional expert consensus). The right panel: AMS 800 consensus commonly uses 61–70 cmH2O. Its fill volumes and pressures belong to that device.
- **ICS16** — ICS AUS consensus 2015, published 2016 (Biardeau et al.) (International consensus, centred on AMS 800). Describes retropubic and alternative balloon pockets and the hazards after pelvic surgery, radiation or mesh.

Next → The third component is the one the patient will touch every day.

30 Pump position and patient access

43:30 – 44:45 · 1:15 · The operation

This slide wants: Place the pump where this patient has already shown he can reach it.

On the slide: A pump position that works for this patient, standing, seated and swollen.

SAY

The pump position is part of the functional result. The patient has already shown us where he can reach, which hand he uses and whether another scrotal device causes confusion. Use that information.

A dependent, accessible position is generally useful, but the exact pocket must suit the patient and device. Too much dissection can encourage migration. Too little space or excessive tension can cause pain, poor access or pressure on adjacent tissue. Keep the pump distinct from the testis and avoid thin, threatened skin.

Before closure, consider how the pump will sit when the patient stands, and whether the tubing will pull it upwards. Check that its orientation permits both normal cycling and access to the deactivation control.

A demonstration pump that is easy to feel on a table does not tell us that a swollen postoperative scrotum will be easy to use. Check accessibility again at activation, with the patient performing the task.

EXAMPLE

Before closure the surgeon tests the pump position with the patient's agreed right-hand side, checks it sits below and distinct from the testis, and pulls gently on the tubing to confirm it will not ride upwards when he stands. At activation, six weeks later, the patient cycles the device himself rather than being shown where it is.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **AP23** — Asia-Pacific AMS 800 consensus 2023 (Chung et al.) (Regional expert consensus). Practical guidance on a dependent, accessible scrotal pump pocket, avoiding migration, tension and confusion with another device.
- **ICS16** — ICS AUS consensus 2015, published 2016 (Biardeau et al.) (International consensus, centred on AMS 800). Operative principle that pump position is part of the functional result and must be rechecked at activation.

- **CCIFU** — Rigicon ContiClassic instructions for use, REV.03, 2023 (Manufacturer labelling). Requires the patient to reach and operate the pump and the deactivation control, which the position must allow.

Next → Then we finish the system, and we hand over the risk with it.

31 System checks and handover

44:45 – 46:00 · 1:15 · The operation

This slide wants: Verify the system before closing, then hand over the risk in writing.

On the slide: Four verifications before closure, and a handover that protects the patient from the next catheter.

SAY

Follow each tube between components. Avoid tension, excessive loops, kinks and entrapment during closure. Keep needles and instruments away from the components.

Prepare and connect the system using its own operative protocol. Use compatible components, remove air as instructed and verify the connections. Cycle the device to check function, then confirm that the cuff is open and the device correctly deactivated. Pressing a button without establishing the cuff state is not enough.

Choose an appropriate small, atraumatically placed catheter for the individual cuff and urethra, and plan early removal after an uncomplicated implant. Device instructions, an injury or a reconstruction may change that plan, and no universal catheter size should be improvised.

The handover should record the model, cuff site and size, balloon specification and position, deactivated state, catheter plan and contact details. The next clinician must know there is an AUS before attempting drainage. The 2026 survey of urologists is the reason we labour this: the gap in deactivation confidence is real.

EXAMPLE

The closing checklist is spoken aloud: tubing traced, connections confirmed, device cycled, cuff open, deactivated. The operation note records model, cuff size and site, balloon pressure and position, "deactivated, cuff open", the catheter plan and the on-call number. The ward is told there is an AUS before any catheter is considered.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **CCIFU** — Rigicon ContiClassic instructions for use, REV.03, 2023 (Manufacturer labelling). Preparation, connection, cycling and deactivation must follow the device's own protocol: steps two and three.
- **ICS16** — ICS AUS consensus 2015, published 2016 (Biardeau et al.) (International consensus, centred on AMS 800). Device-aware handover: record the device, cuff state and catheter plan so the next clinician knows before instrumenting.
- **AP23** — Asia-Pacific AMS 800 consensus 2023 (Chung et al.) (Regional expert consensus). Describes tubing routing, connection checks and the deactivated state at closure, and early catheter removal.
- **CATH13** — Seideman et al. 2013: prolonged catheterisation and erosion (Observational implantation and revision cohort, 200 men). The reason for early removal: catheterisation over 48 hours was associated with erosion in 17 of 44 exposed men.

- **INSTR26** — AUS instrumentation knowledge survey 2026 (Girgin, Neurourol Urodyn) (Self-selected survey of 286 Turkish urologists). The footer: the 2026 survey found gaps in deactivation confidence, which is why step four records deactivation and an urgent contact route.

Next → Let us now see all three spaces together on the model, and follow the tubing between them. Four minutes.

32 UroOps 3D: anatomy and component routes

46:00 – 50:00 · 4:00 · The operation

This slide wants: Follow the anatomy on the model, then follow the tubing between the three spaces.

On the slide: Four minutes on the UroOps 3D model: cuff and dorsal plane, pump access, balloon pocket, tubing route.

SAY

Open uroops3d.com/lab/aus and use the scene controls. Say aloud that this is a teaching schematic with clinical review pending; it shows spatial relationships, not validated anatomy, perfusion or operative instructions.

00:00–00:20: Transition to anatomical relationships. 00:20–01:05: Cuff site view. Discuss viable bulbar tissue and supporting spongiosum. 01:05–01:45: Surgeon's view. Discuss the dorsal plane and why scarred tissue changes dissection. 01:45–02:20: The three spaces view. Discuss accessible pump position. 02:20–03:10: Space of Retzius view. Discuss how previous surgery changes balloon access; describe alternative pockets verbally if absent. 03:10–03:45: Final configuration view. Trace tubing without claiming the schematic proves tension or perfusion. 03:45–04:00: Return to postoperative care.

If loading fails, keep this slide displayed and use the spoken fallback below. Do not spend the allocated session troubleshooting beyond the opening twenty seconds.

FALLBACK 00:00–00:20: "The model is not loading, so let us walk through the same anatomy verbally. Picture the bulbar urethra in the perineum, the pump in the scrotum and the balloon beneath the abdominal wall. We will connect each position to one operative decision."

FALLBACK 00:20–01:05: "Begin with the cuff. It surrounds the urethra and its supporting spongiosal tissue. Our task is to identify a viable segment and preserve that support while creating only the space the cuff needs. A thin, scarred or previously eroded segment does not become healthy because it accepts a measuring tape. The operative report from previous reconstruction is therefore part of the anatomical assessment. Before measuring, release traction and check that the tape encloses the tissue the cuff will actually surround."

FALLBACK 01:05–01:45: "Dorsally, the urethra lies closely related to the corporal bodies. In a normal plane, controlled dissection separates the structures. In a scarred plane, blind force may cross the urethral wall. The decision is to improve exposure, choose another strategy or stage the operation. Transcorporal placement may avoid a particular dangerous plane, but its comparative protection against erosion remains uncertain. It also changes the tissue

within the cuff and the implications for future corporal surgery."

FALLBACK 01:45–02:20: "Next, put the pump in the patient's hand, mentally. Can he find it, distinguish it from the testis and reach its controls? The position must work when sitting as well as standing. An oversized pocket permits migration, while tension and crowding cause discomfort. This is why pump access is agreed before anaesthesia. At activation, let him demonstrate the full operation rather than merely identify where the pump sits."

FALLBACK 02:20–03:10: "Now consider the balloon. After pelvic surgery, the retropubic space may not behave like the textbook space. Adhesions, mesh, bowel, bladder and vessels change the route. An alternative submuscular pocket may be useful, but it must be deliberately and safely created. The choice of pocket and the choice of pressure are separate decisions. Neither a new pocket nor a higher pressure compensates for a poorly selected cuff site. Unexpected resistance during access should prompt reassessment, not force. Confirm the actual product's placement and preparation instructions before committing to the pocket."

FALLBACK 03:10–04:00: "Trace the tubing between all three components. Avoid tension, sharp turns and entrapment under closure. Recheck connections, cycle the system and confirm an open cuff with correct deactivation. The same route helps us investigate a later failure: patient operation, cuff and urethra, tubing, then balloon and fluid system. Future instrumentation has to pass through the cuffed segment, so deactivation and appropriate instrument selection remain essential even when the hydraulics work normally. That is the anatomical logic we wanted the model to demonstrate. Let us return to postoperative care."

EXAMPLE

On the model, start at the cuff: point to the bulbar urethra and the spongiosal tissue around it, then rotate to the dorsal plane and show why a scarred plane changes the dissection. Move to the scrotum and ask the room whether the pump would be reachable seated. Finish in the space of Retzius and trace the tubing back. If the page fails, the spoken fallback in the notes covers the same four stops.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **ICS16** — ICS AUS consensus 2015, published 2016 (Biardeau et al.) (International consensus, centred on AMS 800). Frames the four stops: cuff site and spongiosum, pump access, balloon pocket and tubing route are the consensus operative principles.
- **AP23** — Asia-Pacific AMS 800 consensus 2023 (Chung et al.) (Regional expert consensus). Supplies the practical detail for each space, including alternative balloon pockets and pump pocket limits.
- **CCIFU** — Rigicon ContiClassic instructions for use, REV.03, 2023 (Manufacturer labelling). Reminds us that the model shows relationships only; component placement and preparation follow the device instructions.

Next → Back from the model: the operation is finished, and the first weeks decide whether it was worth doing.

After the implant

50:00 – 60:00 · 10 minutes · teaching slides 33 to 38

Section slide, no clock time. Read the title, then say: "Hand over the risk, diagnose late leakage before revising, and say where the evidence stops."

33 Aftercare, activation and urgent review

50:00 – 51:00 · 1:00 · After the implant

This slide wants: Plan the first six weeks and name the symptoms that bring the patient back early.

On the slide: The planned aftercare, and the symptoms that bring the patient back early.

SAY

After uncomplicated implantation, remove the catheter early and check voiding. Persistent retention needs assessment of deactivation, oedema, cuff fit and obstruction, with safe drainage and no repeated blind catheter attempts. Selected day-case pathways are feasible: the BAUS 2026 audit discharged 88 per cent the same day, and the 2026 meta-analysis found no significant excess of retention or emergency visits, with wide intervals. That supports a selected pathway with verified emptying, not a universal instruction.

Worsening pain, erythema, discharge, fever or voiding difficulty needs specialist review. Suspected erosion requires device-aware investigation, and established infection or erosion generally needs explantation, not antibiotics alone.

ContiClassic specifies four to six weeks deactivated, until pain and oedema settle. At activation, verify patient operation and emptying. Arrange subsequent functional review and access for problems.

Give him the device card and clear instructions to alert clinicians before catheterisation or endoscopy. Safe future care depends on that information reaching the person holding the instrument.

EXAMPLE

A day-case patient voids 300 mL with a 40 mL residual before discharge, carries a device card, and is told to ring if he cannot void, has fever or scrotal redness. He is seen at six weeks for activation, where he cycles the pump himself. That is a selected day-case pathway with verified emptying, which is what the 2026 data support.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **BAUS25** — BAUS consensus 2025 (Bhatt et al., BJU Int) (UK expert consensus after literature review and modified Delphi). Supports a defined postoperative pathway with early catheter removal, verified voiding and clear urgent-review criteria.
- **CCIFU** — Rigicon ContiClassic instructions for use, REV.03, 2023 (Manufacturer labelling). The "4–6 weeks deactivated" line, with activation that includes patient demonstration and device information.
- **AUA24** — AUA/GURS/SUFU incontinence after prostate treatment, amended 2024 (US guideline). Supports device cards, lifetime counselling and specialist review of suspected infection or erosion rather

than antibiotics alone.

- **CATH13** — Seideman et al. 2013: prolonged catheterisation and erosion (Observational implantation and revision cohort, 200 men). Supports removing the catheter early after an uncomplicated implant, because prolonged urethral catheterisation was associated with erosion.
- **BAUSDAY26** — BAUS 2026 abstract P4-8: day-case AUS audit (Conference audit, 98 men). The footer's "day-case audit": 88% same-day discharge in 98 men in an established service; selected and short follow-up.
- **DAYMETA26** — Day-case AUS meta-analysis 2026 (Grigoryan et al.) (Systematic review, six studies, 2,424 men; four pooled). Six studies, 2,424 men: no significant excess of retention (RR 1.44, CI 0.81–2.56) or 90-day emergency visits with day-case care, with serious bias and wide intervals.
- **DAYCASE26** — Day-case versus inpatient AUS 2026 (Youssef et al.) (Retrospective single-centre comparison, 177 men). 177 men: 30-day complications 11.3% ambulatory versus 8.2% inpatient, p = 0.78, with selection by ASA and social factors. A non-significant result, not proof of equal safety.

Next → Our last patient did everything right and was dry for three years. Then he was not.

34 Case 4: leakage after three good years

51:00 – 52:30 · 1:30 · After the implant

This slide wants: Diagnose late leakage before choosing a revision.

On the slide: Case 4: late leakage is a diagnosis to make, not a cuff size to shrink.

SAY

Our fourth fictional patient had three useful years with his AUS and now leaks again. Would you book a smaller cuff? [Pause for five seconds.]

Start by separating persistent leakage since activation from true recurrence after a successful interval. Ask whether this is activity-related leakage, urgency, overflow or difficulty cycling. Observe his technique and check the device state.

Review urine and residual, examine the components, and use appropriately deactivated cystoscopy to look for erosion and assess the urethra. Investigate fluid loss, mechanical problems and bladder dysfunction as indicated.

Urethral atrophy is an explanation to establish, not a diagnosis made by the passage of time. Bergeson's revision series attributed only eight per cent of failures to atrophy and a third to the pressure-regulating balloon, and the 2026 laboratory study of retrieved balloons shows why: they stiffen and thin with age. That is a mechanism, not an instruction to replace every balloon at three years. Capsular restriction and poor initial sizing can also complicate the label.

The answer may be retraining, bladder treatment, component revision, a revised cuff strategy or explantation. In an older system, discuss whole-system replacement. The French national cohort reminds us that second implants have 68 per cent reintervention-free survival at five years, and that only a fifth of men whose first device was removed ever received another. Repeated failure or a compromised urethra warrants specialist review before another tissue-consuming procedure.

EXAMPLE

Case 4 is on the slide. A good answer sounds like this: "Before any revision I would watch him cycle the pump, check urine and residual, examine the components, and do a cystoscopy with the device deactivated to look for erosion. I would check the balloon and fluid system, because a third of revisions in one series were balloon failures. Atrophy is the diagnosis left over, not the first guess."

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **ATROPHY20** — Bergeson et al. 2020: causes of AUS revision (Selected contemporary revision cohort, 137 revisions, erosions excluded). Why atrophy is the diagnosis left over: in 137 revisions, atrophy was assigned in 8% and balloon failure in 34%, so mechanics are checked first.
- **BAUS25** — BAUS consensus 2025 (Bhatt et al., BJU Int) (UK expert consensus after literature review and modified Delphi). Supports a structured assessment of recurrent leakage, including device-aware cystoscopy, before revision.
- **AUA24** — AUA/GURS/SUFU incontinence after prostate treatment, amended 2024 (US guideline). Supports evaluating patient technique, erosion, mechanical failure and bladder dysfunction before choosing revision, and counselling about repeat surgery.
- **CCIFU** — Rigicon ContiClassic instructions for use, REV.03, 2023 (Manufacturer labelling). Requires the cuff deflated and the device deactivated for the cystoscopy, and describes the device states to check.
- **AUASAFER26** — AUA 2026 abstract IP09-17: SAFER erosion management (Conference abstract, title and programme verified only). A 2026 abstract comparing in situ urethroplasty with conservative management at explantation for erosion; results unavailable, so no change to practice.
- **PRBFAT26** — Pressure-regulating balloon fatigue 2026 (Salvino et al., Urology) (Laboratory study of 32 retrieved balloons). The footer's "PRB fatigue study": 32 retrieved balloons stiffened and thinned with age, with microcracks in the oldest. A mechanism, not an instruction to replace balloons at three years.
- **EROSION26** — Immediate urethral repair at explantation 2026 (Park et al., World J Urol) (Selected single-centre series, 16 men, median follow-up 14 months). Immediate urethral repair at explantation in 16 men: one stricture but seven dehiscences; no comparator, so no universal repair policy.
- **SECOND26** — Second AUS survival, national French cohort 2026 (Tokarski et al.) (National administrative cohort, 1,619 second implants). For the revision conversation: 1,619 second implants had 68% reintervention-free survival at five years, and only 21% of men whose first device was removed received another.

Next → So where does all this leave us? Here is what we think is established, and what honestly is not.

35 Established benefit and unresolved questions

52:30 – 53:15 · 0:45 · After the implant

This slide wants: Separate what is established from what is plausible and what is still open.

On the slide: What is supported, what is plausible, and what remains open.

SAY

AUS can deliver substantial benefit, with revision and erosion part of its lifetime burden. Tissue preservation, contamination control and planned instrumentation have strong rationale, although individual technical details often lack comparative evidence.

We cannot name one best salvage operation for every fragile urethra, one optimal irrigant or a proven universal preparation duration. New device features, including the first electronic sphincters, need outcome evidence with comparable populations and definitions before engineering advantages become clinical claims.

Make that uncertainty visible in consent and operative planning.

EXAMPLE

If asked "which is better, ContiClassic or AMS 800?", the supported answer is: "Selected patients gain substantial benefit from an AUS. There is no comparative long-term ContiClassic outcome data, and the electronic sphincter has six patients at one year. The question is open, and I will not close it from a device survival figure."

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **MASTER22** — MASTER trial 2022 (Constable et al., NIHR HTA) (UK multicentre randomised non-inferiority trial, 380 men, sling versus AUS). The "established benefit" column: in a randomised trial both arms improved substantially, and AUS led on several secondary outcomes.
- **TC20** — Redmond et al. 2020: transcorporal placement in fragile urethras (Retrospective series, 76 patients, 24 transcorporal). With TC23, the "best salvage strategy" uncertainty: one fragile-urethra series favoured transcorporal placement numerically.
- **TC23** — Kurtzman et al. 2023: high-risk transcorporal AUS (Retrospective single-surgeon high-risk cohort, 68 men, 77 cuffs). The other direction: more explants after transcorporal placement in a high-risk cohort. Together they leave the salvage question open.
- **PREP24** — Bourgi et al. 2024: AMS 800 skin-preparation protocols (Retrospective sequential-era comparison, 156 procedures). The "optimal preparation duration" uncertainty: the only AUS preparation comparison showed no difference and never tested fifteen minutes.
- **CCSAFETY** — Wilson et al.: first ContiClassic safety outcomes, IJIR 2024 (Uncontrolled early device series, 116 implants, mixed indications). The "comparable long-term ContiClassic outcomes" gap: one early series, 116 implants, twelve months, no continence endpoint.
- **DEVICE26** — Emerging AUS devices review 2026 (Schnell et al., Neurourol Urodyn) (Narrative review). The footer's "2026 reviews": new designs described without direct comparisons or long-term outcomes.
- **UROACTIVE26** — UroActive electronic AUS first-in-human 2026 (Mozer et al., J Urol) (Single-arm feasibility study, six men). The footer's "early electronic-AUS data": six men, all activated, no revision at one year, nine remote-control deficiencies. Zero revisions in six men is not zero risk.

Next → Let us condense the hour into ten decisions. Five belong before theatre.

36 Five decisions before implantation

53:15 – 54:00 · 0:45 · After the implant

This slide wants: Carry five pre-theatre decisions out of the room.

On the slide: The five pre-theatre decisions in one list.

SAY

The first five take-home messages belong before theatre. Establish the leakage mechanism and the improvement that matters. Watch the patient use a pump and assess access over time. Choose investigations to answer explicit questions, acknowledging the differences between guidelines.

Keep detrusor overactivity, underactivity and poor compliance separate, because they lead to different decisions. Finally, establish a stable outlet and think through future instrumentation.

A successful implant starts with a patient, bladder and urethra that can support the intended result. Apply them to the cases: optimise case one, investigate case two, and stage a reconstructive plan for case three.

EXAMPLE

Apply the five to the cases: case 1 fails decision four (urgency not yet separated, so optimise); case 2 fails decision three (emptying unresolved, so investigate); case 3 fails decision five (no stable outlet, so stage). Each case is turned down by a named decision, not by a feeling.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **BAUS25** — BAUS consensus 2025 (Bhatt et al., BJU Int) (UK expert consensus after literature review and modified Delphi). Decisions one, three and five: demonstrated SUI, investigations in a pathway, and a stable outlet before surgery.
- **EAU26** — EAU non-neurogenic male LUTS guideline 2026 (European guideline). Decision three: structured assessment with diary, pad weight and residual before invasive treatment.
- **AUA24** — AUA/GURS/SUFU incontinence after prostate treatment, amended 2024 (US guideline). Decisions one and two: bothersome demonstrated SUI, a patient who can use the device, and counselling about the lifetime burden.
- **DU23** — Han et al. 2023: AUS with detrusor underactivity (Retrospective cohort, 78 patients, 55 with BCI below 100). Decision four: detrusor underactivity is a separate problem with selected, limited data, not an automatic exclusion or an automatic clearance.
- **ICS16** — ICS AUS consensus 2015, published 2016 (Biardeau et al.) (International consensus, centred on AMS 800). Decisions two and four: dexterity and cognition, and bladder evaluation separating storage from emptying.

Next → And five decisions protect the result once the implant is open.

37 Five decisions protecting the result

54:00 – 55:00 · 1:00 · After the implant

This slide wants: Carry five result-protecting decisions out of the room.

On the slide: The five decisions that protect the result, from preparation to revision.

SAY

Preparation must be correct for the product, anatomy and implant pathway. A fifteen-minute ritual is not a substitute. In theatre, preserve the tissue you need, measure what the cuff will actually enclose, and correct uncertainty before closing.

Use the instructions for the device being implanted. Similar components do not make pressure ranges, preparation or operating sequences interchangeable. Recognised injury or unsafe contamination is a reason to change the plan.

Confirm the cuff state, drainage and the patient's ability to use the device. Make future instrumentation precautions explicit.

When leakage returns, diagnose it before selecting a smaller cuff or higher pressure. The goal is useful continence over time, with safe storage, emptying and an agreed plan for further treatment.

EXAMPLE

Apply the five to case 4: decision ten says diagnose before revising. Apply them to the fifteen-minute wash: decision six says correct antisepsis for the product, not a duration. Apply them to the dropped cuff: decision eight says explicit stop conditions, so the cuff is discarded.

STUDIES ON THIS SLIDE: WHAT EACH ONE DOES HERE

- **CCIFU** — Rigicon ContiClassic instructions for use, REV.03, 2023 (Manufacturer labelling). Decision eight: the actual device protocol for preparation, pressure, filling and deactivation.
- **NICE** — NICE NG125: surgical site infection (UK national guidance). Decision six: correct antisepsis, drying and no irrigation purely to prevent SSI.
- **AP23** — Asia-Pacific AMS 800 consensus 2023 (Chung et al.) (Regional expert consensus). Decisions seven and nine: tissue preservation, sizing without traction, system checks and a deactivated handover.
- **ATROPHY20** — Bergeson et al. 2020: causes of AUS revision (Selected contemporary revision cohort, 137 revisions, erosions excluded). Decision ten: a third of revisions were balloon failures and few were atrophy, so diagnose before revising.
- **BAUS25** — BAUS consensus 2025 (Bhatt et al., BJU Int) (UK expert consensus after literature review and modified Delphi). Decisions six and nine: a defined perioperative pathway and handover with urgent-review criteria.

Next → That is our ten. The last five minutes are yours.

38 Discussion

55:00 – 60:00 · 5:00 · After the implant

This slide wants: Run five minutes of discussion on decisions, not preferences.

On the slide: Five minutes of moderated discussion on decisions, not preferences.

SAY

55:00–55:20: "We would like to use these final five minutes on decisions rather than preferences alone. Choose one of the cases, tell us what you would do, and identify the finding that would make you change your plan."

55:20–57:20: Invite a response on bladder dysfunction or urethral stability. Take one contrasting view and a brief reply. Ask, if needed: "Is that a guideline requirement, a device instruction, or your interpretation of the evidence?"

57:20–59:40: Invite a second question on operative practice, preparation or failure. Use the prepared answers in the presenter guide if relevant. Keep individual contributions brief enough to allow disagreement and clarification.

59:40–60:00: "Thank you. The useful differences are now clear: what the evidence supports, what the device requires and where our judgement remains decisive. Those distinctions should be visible in our consent, our operation notes and the outcomes we report."

A show of hands is audience opinion; do not convert it into consensus evidence. For a question beyond the retrieved evidence, state the uncertainty precisely.

EXAMPLE

A questioner says "we always do urodynamics". The moderating question is: "Is that a guideline requirement, a device instruction, or your interpretation of the evidence?" If the room splits, that split is the useful result; a show of hands is opinion, not evidence.

Backup slides, outside the clock

- **Outcome evidence: device and endpoint.** Three studies, three different endpoints: AUSCO 2026 (pad outcomes at twelve months), Linder 2015 (freedom from secondary surgery over ten years) and Wilson 2024 (early ContiClassic device survival). Read the denominators aloud; they do not rank devices.
- **Comparative and preparation evidence.** MASTER 2022/2026 (randomised, non-inferior at 24 months, further surgery 11% versus 2%), Yeung 2013 (skin-culture surrogate, 8% versus 32%) and the 2026 gentamicin-withdrawal cohort (three infections in 173 procedures; not equivalence).
- **Evidence gaps.** Seven open comparative questions from section 7 of the manuscript.
- **UroOps 3D lesson: recorded run-through** (animated deck only). 102 seconds, no sound, click to play. Use only if the live page fails.
- **Sources 1–3.** All 58 citation codes with live links.

Source glossary: what each source is and what it says

Every code used in the deck, in the order of the reference slides. The per-slide lines above say what each source does on that slide; this list gives the whole finding and its limit.

- **BAUS25** — BAUS consensus 2025 (Bhatt et al., BJU Int). *What it is:* UK expert consensus after literature review and modified Delphi. *What it says:* Sets the UK assessment and perioperative pathway: VUDS for surgical candidates, severity-dependent recovery periods (severity graded by 24-hour pad weight on the Nitti scale: mild under 100 g, moderate 100 to 400 g, severe over 400 g), treatment of narrowing before continence surgery, and a five-minute topical field scrub that it labels as Delphi extrapolation. Consensus supports a consistent pathway; it does not prove each interval or step is uniquely optimal.
- **EAU26** — EAU non-neurogenic male LUTS guideline 2026. *What it is:* European guideline. *What it says:* Recommends structured symptom assessment, diary, pad-weight and residual measures; weakly recommends invasive urodynamics before invasive treatment while acknowledging limited outcome evidence. A weak recommendation signals uncertainty, not a valueless test.
- **AUA24** — AUA/GURS/SUFU incontinence after prostate treatment, amended 2024. *What it is:* US guideline. *What it says:* Permits surgery from six months for bothersome non-improving SUI and recommends offering it at one year; cystourethroscopy before SUI surgery is expert opinion; urodynamics is conditional (Grade C) when it helps diagnosis or counselling. It also supports pump suitability, lifetime counselling and interval cystoscopy at least four to six weeks after treating BNC or VUAS.
- **EAUSTR26** — EAU urethral strictures guideline 2026. *What it is:* European guideline. *What it says:* Distinguishes anterior strictures from posterior stenosis and gives reconstruction-specific pathways, including deferring continence surgery for three to six months after redo vesicourethral anastomosis. That interval belongs to that setting, not to every dilation or urethroplasty.
- **ICS16** — ICS AUS consensus 2015, published 2016 (Biardeau et al.). *What it is:* International consensus, centred on AMS 800. *What it says:* Covers assessment, operative principles, complications and aftercare: adequate dexterity and cognition, bladder and outlet evaluation, and device-aware instrumentation. Its technical and catheter advice is AMS-era and context-specific, not a universal ContiClassic rule.
- **AP23** — Asia-Pacific AMS 800 consensus 2023 (Chung et al.). *What it is:* Regional expert consensus. *What it says:* Practical consensus on selection, investigation, troubleshooting and aftercare, including common AMS 800 pressure selection (61–70 cmH2O) and preparation. Its fill volumes and preparation sequence belong to AMS 800, not automatically to ContiClassic.
- **CCIFU** — Rigicon ContiClassic instructions for use, REV.03, 2023. *What it is:* Manufacturer labelling. *What it says:* Requires associated urge, overflow or instability to be treated and controlled or resolved, contraindicates irresolvable instability, warns about poor compliance and thin spongiosum, requires cuff deflation and deactivation before instrumentation, specifies four to six weeks deactivated, and describes a longer postoperative antibiotic regimen than stewardship sources. These are product instructions, not proof of efficacy. Country and lot applicability and the separate OR protocol remain unverified.
- **ICSUDS17** — ICS good urodynamic practices 2017 (Rosier et al.). *What it is:* Testing standard. *What it says:* Sets quality standards for uroflowmetry, cystometry and pressure-flow testing. It tells us to confirm a trace is credible and representative before labelling DO, DU, compliance or obstruction. It sets no AUS eligibility threshold.
- **AUAOAB24** — AUA/SUFU idiopathic OAB guideline 2024 (Cameron et al.). *What it is:* US guideline. *What it says:* Evidence-based behavioural, drug and procedural options for OAB with shared decision-making, including the retention implications of botulinum toxin. It does not establish an optimal sequence around AUS implantation.

- **NICE** — NICE NG125: surgical site infection. *What it is:* UK national guidance. *What it says:* Soap washing before surgery, appropriate antiseptic skin preparation with drying, clipping rather than shaving if hair removal is needed, and no wound irrigation solely to reduce SSI. It tests no fifteen-minute AUS preparation.
- **WHO09** — WHO hand hygiene guidance 2009. *What it is:* International standard. *What it says:* Covers surgical hand antisepsis. Its relevance is the distinction between preparing the team's hands and preparing the patient's skin. It sets no patient-field scrub duration.
- **CHLORA26** — ChloraPrep Swabstick UK product information, 2026. *What it is:* Product information (SmPC). *What it says:* Permitted sites, allergy, intact-skin-only use, a 30-second application with complete air drying and flammability warnings for that presentation. It shows why the actual formulation matters; it is not a universal genital preparation prescription.
- **IDSA19** — IDSA asymptomatic bacteriuria guideline 2019 (Nicolle et al.). *What it is:* US infectious-disease guideline. *What it says:* Suggests against screening for or treating asymptomatic bacteriuria solely for AUS or penile prosthesis implantation, with standard prophylaxis retained. The recommendation is weak, on very-low-quality evidence, and differs from the BAUS urine pathway.
- **AUAABX20** — AUA antimicrobial prophylaxis statement 2020 (Lightner et al.). *What it is:* Consensus statement informed by literature review. *What it says:* Correct timing, redosing and the shortest necessary duration; single-dose prophylaxis for most urological procedures and no unnecessary continuation. It does not settle every high-risk implant or override product precautions.
- **CCSAFETY** — Wilson et al.: first ContiClassic safety outcomes, IJIR 2024. *What it is:* Uncontrolled early device series, 116 implants, mixed indications. *What it says:* Eight revisions (6.9%), no reported infections and 93.2% twelve-month Kaplan–Meier device survival; four revisions were attributed to sizing errors. Device survival is not continence, there is no comparator, and only abstract-level results were accessible.
- **MASTER22** — MASTER trial 2022 (Constable et al., NIHR HTA). *What it is:* UK multicentre randomised non-inferiority trial, 380 men, sling versus AUS. *What it says:* At twelve months the sling met non-inferiority on the patient-reported continence endpoint (difference −3.4 points, 95% CI −11.7 to +4.8, 15-point margin). Both groups improved and several secondary outcomes favoured AUS. Further continence surgery within twelve months: 13 sling-arm men (12 went on to an AUS, one to a second sling) versus three AUS-arm men (revisions for mechanical failure: reservoir change, balloon replacement or repositioning). Prior implants and unresolved narrowing were excluded, so it says nothing about complex revision practice.
- **UDS09** — Lai et al. 2009: urodynamics before AUS. *What it is:* Selected cohort, 129 first AUS after prostatectomy, mean follow-up 34 months. *What it says:* Adverse preoperative urodynamic findings did not significantly worsen later pad use, but the authors warned that a good continence result with poor compliance can still expose the upper tracts to high storage pressure. It separates pad outcomes from bladder safety; it does not say abnormal UDS can be ignored.
- **OAB11** — Lai and Boone 2011: OAB and post-prostatectomy AUS. *What it is:* Selected cohort, 129 first AUS after prostatectomy. *What it says:* Preoperative OAB did not worsen overall continence, but 71% with mixed symptoms still had OAB afterwards and 23% with pure SUI developed new OAB. Stress leakage can improve while urgency persists.
- **DO24** — Bhatt et al. 2024: preoperative DO systematic review. *What it is:* Systematic review, 11 studies, 792 patients, AUS and sling. *What it says:* Did not find a statistically significant overall excess of failure sufficient to deny surgery solely because DO was present, but definitions and reporting were heterogeneous. It supports counselling about persistent storage symptoms; it does not promise urgency resolves.
- **DOFAIL23** — Krughoff and Peterson 2023: earlier AUS reintervention. *What it is:* Single-centre analysis of 92 men already undergoing reintervention. *What it says:* DO was associated with earlier reintervention (adjusted HR 1.95, 95% CI 1.22–3.1) within that selected group. It tempers reassurance; it cannot give the failure probability for all primary recipients.

- **DU23** — Han et al. 2023: AUS with detrusor underactivity. *What it is:* Retrospective cohort, 78 patients, 55 with BCI below 100. *What it says:* Postoperative residuals did not differ significantly by DU status, although flow stayed lower. It excluded complications needing revision within three months, which weakens the safety inference. BCI below 100 is the study definition, not an eligibility cutoff.
- **CIC23** — Krughoff et al. 2023: intermittent catheterisation after bulbar AUS. *What it is:* Retrospective selected cohort, 57 men with CIC history. *What it says:* Eighteen continued or started CIC after AUS; no statistically significant difference in erosion or later removal was found. Small numbers and selection mean a missed difference is possible. Feasible in carefully planned men; not equivalence to men who never catheterise.
- **CATH13** — Seideman et al. 2013: prolonged catheterisation and erosion. *What it is:* Observational implantation and revision cohort, 200 men. *What it says:* Urethral catheterisation for more than 48 hours was associated with erosion (17 of 44 exposed men), alongside radiation and revision. It supports avoiding unnecessary prolonged urethral access; 48 hours is not a guaranteed safe boundary.
- **UDS14** — Weissbart et al. 2014: absent leak during intubated urodynamics. *What it is:* Cohort, 169 men. *What it says:* Thirty-two men did not leak with the urethral test catheter in place and all leaked once it was removed, with useful later AUS outcomes. A catheter can mask real stress leakage; this does not justify skipping clinical demonstration.
- **RT22** — Zhang and Xu 2022: radiation meta-analysis. *What it is:* Meta-analysis of 18 observational studies. *What it says:* Prior radiation was associated with lower odds of complete continence and more erosion or explantation (pooled erosion OR about 2.0, 95% CI 1.15–3.45). An odds ratio is not a personal probability, and radiation is a counselling and tissue-planning factor, not an automatic exclusion.
- **URETH26** — Davis et al. 2026: AUS after urethroplasty. *What it is:* Retrospective multicentre series, 178 cases, 15 institutions. *What it says:* Complications requiring explantation occurred in 56.2% after transecting versus 23.5% after non-transecting urethroplasty. Selection, stricture severity and follow-up confound the comparison; it supports a detailed reconstructive history, not a claim that every transection was avoidable.
- **TC20** — Redmond et al. 2020: transcorporal placement in fragile urethras. *What it is:* Retrospective series, 76 patients, 24 transcorporal. *What it says:* Revision 20.8% versus 36.5% and erosion 8.3% versus 17.3%, numerically favouring transcorporal placement, but the erosion difference was not statistically significant and allocation was not randomised.
- **TC23** — Kurtzman et al. 2023: high-risk transcorporal AUS. *What it is:* Retrospective single-surgeon high-risk cohort, 68 men, 77 cuffs. *What it says:* Transcorporal placement was associated with more infection or erosion-related explants (HR 2.74). Surgeons may have chosen it for the worst tissue, so it cannot prove the technique caused harm. With Redmond it blocks any confident claim of erosion protection.
- **ATROPHY20** — Bergeson et al. 2020: causes of AUS revision. *What it is:* Selected contemporary revision cohort, 137 revisions, erosions excluded. *What it says:* Atrophy was assigned in 8% and pressure-regulating balloon failure in 34%. It tells us to check mechanics before labelling late leakage as atrophy; it does not show atrophy never occurs.
- **PREP24** — Bourgi et al. 2024: AMS 800 skin-preparation protocols. *What it is:* Retrospective sequential-era comparison, 156 procedures. *What it says:* Intensified povidone preparation showed no association with device removal (OR 0.97, 95% CI 0.28–3.37). Wide interval, era confounding and inconsistent reporting; it never compared fifteen minutes against a shorter preparation.
- **YEUNG13** — Yeung et al. 2013: GU prosthetic skin preparation RCT. *What it is:* Single-centre randomised trial, 100 initial GU prosthetic procedures, 50 per arm. *What it says:* Post-preparation skin cultures were positive in 8% with chlorhexidine-alcohol versus 32% with povidone-iodine (P = 0.0091). A microbiological surrogate in a mixed-device population; it did not show fewer AUS infections and tested no duration.
- **ABX18** — Adamsky et al. 2018: postoperative oral antibiotics. *What it is:* Claims-based observational study, 3,594 AUS recipients. *What it says:* Explantation within three months was 3.9% with and 4.0% without

postoperative oral antibiotics, with no protective association after adjustment. It challenges routine prolonged oral prophylaxis; it cannot prove identical safety in every subgroup.

- **TOUCH23** — Ziegelmann et al. 2023: minimal-touch AUS technique. *What it is:* Historical comparison of a contamination-control bundle. *What it says:* Crude infections appeared lower after the bundle, but twelve-month cumulative incidence differences were not statistically significant. A sensible bundle to study; it cannot credit one glove change, drape or pause.
- **LINDER15** — Linder et al. 2015: long-term primary AMS 800 outcomes. *What it is:* Historical specialist cohort, 1,082 primary implants, median follow-up 4.1 years. *What it says:* Kaplan–Meier freedom from any secondary surgery was 74% at five years and 57% at ten years. This is maintenance and revision counselling, not pad-free continence.
- **AUSCO26** — AUSCO study 2026 (Kaufman et al.): prospective AMS 800 outcomes. *What it is:* Prospective single-arm, manufacturer-funded, 17 sites, 115 primary implants. *What it says:* At twelve months after activation 91 of 97 had at least 50% pad-weight reduction and 61 of 101 reported zero pads (60.4%); nine participants had revisions. Contemporary counselling data with missing denominators, expert centres and one year of follow-up; no device comparison.
- **AUASTATUS26** — AUA current guideline register, July 2026. *What it is:* Official register page. *What it says:* Still lists Incontinence after Prostate Treatment as amended in 2024; no 2026 amendment exists. Meeting abstracts and update courses are not a new guideline.
- **BAUSDAY26** — BAUS 2026 abstract P4-8: day-case AUS audit. *What it is:* Conference audit, 98 men. *What it says:* 88% same-day discharge in an established service. Short follow-up and a selected population; it does not establish long-term equivalence or antibiotic policy.
- **BAUSSCI26** — BAUS 2026 abstract P12-7: bulbar AUS in spinal cord injury. *What it is:* Small retrospective conference series. *What it says:* Explored predominantly bulbar placement in SCI. Small, selected, no bladder-neck comparator; a specialist discussion point that keeps safe storage and an emptying plan central.
- **AUAUDS26** — AUA 2026 abstract IP09-22: pre-AUS urodynamics and cystoscopy. *What it is:* Conference abstract, title and programme verified only. *What it says:* Revisited the value of preoperative UDS and cystoscopy. Full methods and results were unavailable, so no numerical result or change in practice is inferred.
- **AUAABX26** — AUA 2026 abstract IP09-18: postoperative antibiotics. *What it is:* Conference abstract, title and programme verified only. *What it says:* Its title states no infection benefit from postoperative antibiotics. Full methods and results were unavailable; nothing numerical is inferred.
- **AUASAFER26** — AUA 2026 abstract IP09-17: SAFER erosion management. *What it is:* Conference abstract, title and programme verified only. *What it says:* Compared in situ urethroplasty with conservative management at explantation for cuff erosion. Results were unavailable; no practice change is inferred.
- **MASTER26** — MASTER 24-month results 2026 (Constable et al., Eur Urol Focus). *What it is:* Extended follow-up of the same 380 randomised men. *What it says:* Sling remained non-inferior on the strict self-reported continence endpoint at 24 months (difference −0.6 points, 95% CI −9.2 to +8.0). Further surgery: 11% after sling versus 2% after AUS; secondary outcomes generally favoured AUS. Non-inferiority on one endpoint is not identical effectiveness.
- **URETHMETA26** — Prior urethroplasty meta-analysis 2026 (Deameh et al., Urology). *What it is:* Meta-analysis of four observational studies, 533 men. *What it says:* Prior urethroplasty was associated with higher explantation (RR 2.05, 95% CI 1.08–3.89); the pooled erosion difference was not significant. Transecting versus non-transecting repair was associated with more erosion (RR 2.34) and explantation (RR 2.38). Observational and partly overlapping with Davis; not causal.
- **IRRIG26** — Gentamicin withdrawal cohort 2026 (Skrzypczyk et al., Antibiotics). *What it is:* Single-centre before-after study, 173 uncoated AMS 800 procedures. *What it says:* 90-day SSI 1 of 59 with gentamicin versus 2 of 114 without (difference +0.1 points, 95% CI −7.4 to +4.7). Three events in total, immersion and

irrigation changed together, and broad systemic prophylaxis continued. Not equivalence, and not transferable to coated AMS 800 or ContiClassic.

- **DAYMETA26** — Day-case AUS meta-analysis 2026 (Grigoryan et al.). *What it is:* Systematic review, six studies, 2,424 men; four pooled. *What it says:* No significant difference in retention (RR 1.44, 95% CI 0.81–2.56) or 90-day emergency visits (RR 0.91). Moderate-to-serious bias and wide intervals; same-day discharge is a selected pathway, not an instruction.
- **DAYCASE26** — Day-case versus inpatient AUS 2026 (Youssef et al.). *What it is:* Retrospective single-centre comparison, 177 men. *What it says:* 30-day composite complications 11.3% ambulatory versus 8.2% inpatient, p = 0.78. Inpatient selection depended on ASA and social factors; a non-significant result is not proof of equal safety.
- **FRAIL26** — Frailty and SUI surgery 2026 (Nik-Ahd et al., J Urol). *What it is:* Medicare claims cohort, 7,252 men, AUS and sling. *What it says:* Claims-defined frailty was associated with 30-day complications independently of age and comorbidity (adjusted RR 1.5 prefrail, 2.5 frail) and with revision or removal. Mixed procedures and a claims index; it validates no bedside pump score.
- **PAD26** — Pad count versus 24-hour pad weight 2026 (Domínguez Argomedo et al., BJU Int). *What it is:* Retrospective analysis of prospective follow-up, 2,040 men, 8,787 paired measurements. *What it says:* Pad count tracked leakage well overall (Spearman rho 0.94), but two to four pads a day had considerable weight variability. Measure pad weight when the intermediate count leaves severity uncertain.
- **PRBFAT26** — Pressure-regulating balloon fatigue 2026 (Salvino et al., Urology). *What it is:* Laboratory study of 32 retrieved balloons. *What it says:* Age correlated with stiffness and reduced wall thickness; microcracks appeared in the oldest specimens. Mechanistic evidence from selected retrieved parts; it does not tell us to replace a balloon because three years have passed.
- **REGRET26** — Decisional regret after AUS 2026 (Roth et al., Eur Urol Open Sci). *What it is:* Retrospective single-centre survivor cohort, 91 respondents, median follow-up 82 months. *What it says:* Reoperation was the only independent predictor of regret (+18 points on the Decisional Regret Scale); continence benefit persisted but deteriorated over time. Counsel about residual leakage, future surgery and psychological burden.
- **INSTR26** — AUS instrumentation knowledge survey 2026 (Girgin, Neurourol Urodyn). *What it is:* Self-selected survey of 286 Turkish urologists. *What it says:* Found training and confidence gaps in AUS deactivation and instrumentation. It supports device cards and an urgent-contact route; its proposed catheter sizes and durations cannot become universal rules.
- **EROSION26** — Immediate urethral repair at explantation 2026 (Park et al., World J Urol). *What it is:* Selected single-centre series, 16 men, median follow-up 14 months. *What it says:* One stricture after repair (6.3%) but seven wound dehiscences needing closure (43.8%). No drainage-only comparator; it does not establish universal immediate repair.
- **DEVICE26** — Emerging AUS devices review 2026 (Schnell et al., Neurourol Urodyn). *What it is:* Narrative review. *What it says:* Describes new hydraulic and electronic designs while stressing the lack of robust direct comparisons and long-term outcomes. Engineering advantages are not proven clinical superiority.
- **SECOND26** — Second AUS survival, national French cohort 2026 (Tokarski et al.). *What it is:* National administrative cohort, 1,619 second implants. *What it says:* Reintervention-free survival 68% at five years and 61% at ten years; only 21% of men whose first device was removed received another. It strengthens consent about the path after erosion or explantation.
- **TCSYS25** — Transcorporal placement systematic review, online 2025 (Domínguez Gutiérrez et al.). *What it is:* Systematic review, 20 studies, eight comparative. *What it says:* Transcorporal placement was associated with more revisions (OR 2.99), less erosion (OR 0.35) and less infection (OR 0.33), with no significant explantation difference. Non-randomised, heterogeneous risk selection, and the pooled estimates conflict with some cohorts.

- **RTTIME25** — Radiation and cuff-removal timing cohort 2025 (Lee et al., World J Urol). *What it is:* Retrospective cohort, 283 men. *What it says:* Prior radiation was associated with cuff removal (HR 3.30, 95% CI 1.05–10.40). An exploratory trend towards later implantation after radiation was not significant and validates no waiting rule.
- **UROACTIVE26** — UroActive electronic AUS first-in-human 2026 (Mozer et al., J Urol). *What it is:* Single-arm feasibility study, six men. *What it says:* All activated successfully with no explant or revision in the first year; nine device deficiencies involved the remote control. Zero revisions in six men is not zero risk.
- **REPORT26** — Standardised complication reporting 2026 (Roessler et al., World J Urol). *What it is:* Retrospective series, 227 implants, systematic complication capture. *What it says:* 47% early complications, mostly minor; 7.9% major; five-year explantation-free survival 73%. Includes double and transcorporal cuffs, so it is not a universal primary single-cuff rate. It shows why similar-looking percentages differ.